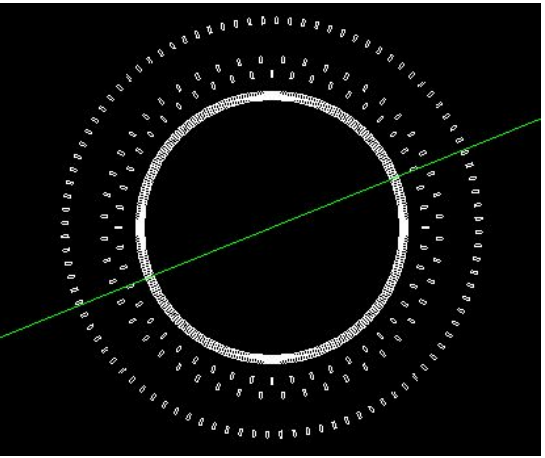
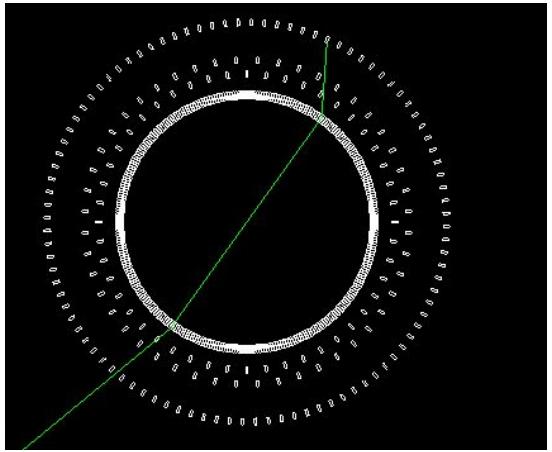
Date: 18/08/25

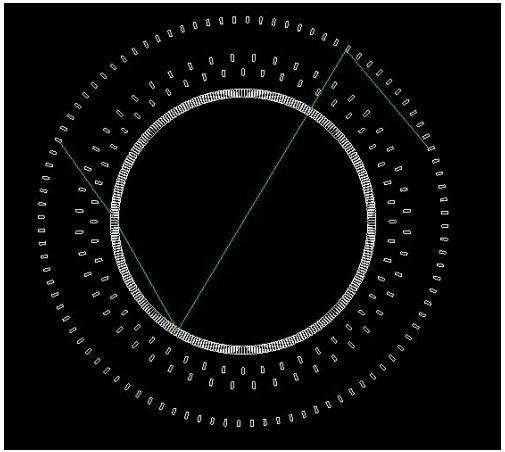
Starting on polarisation studies in two photon case

Added the another photon gun

2nd photon gun at the same position which shoots 511 keV photons in the opposite direction to the first one and has polarization perpendicular to the polarization vector of the first photon.



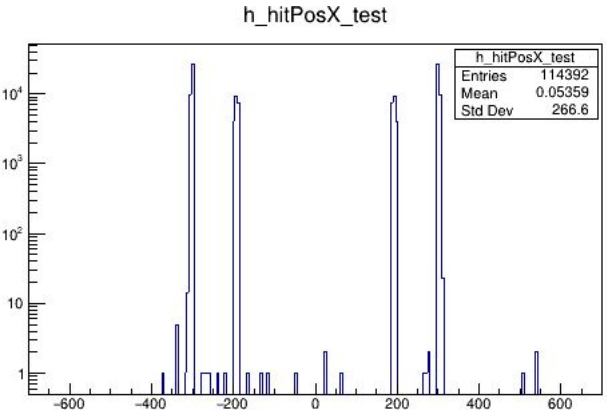


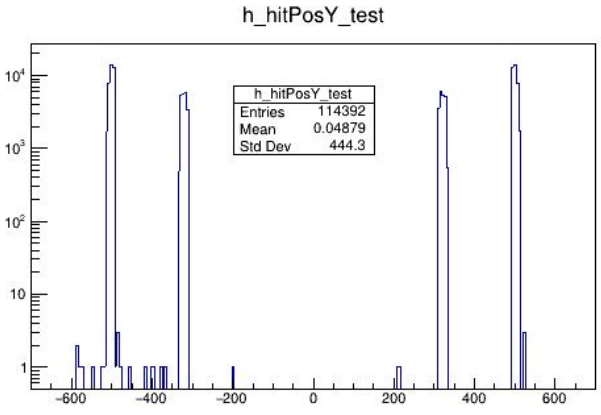


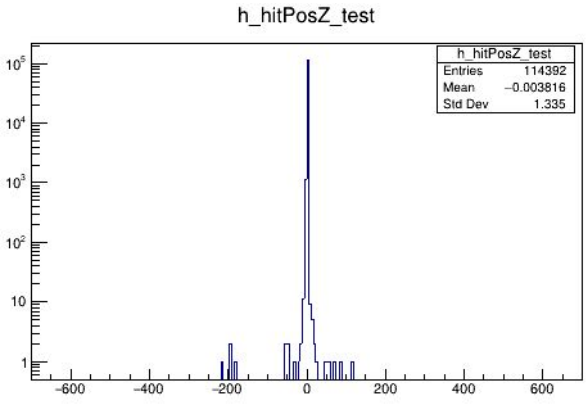
generator.cc

Distribution of hit positions

Date: 19/08/25

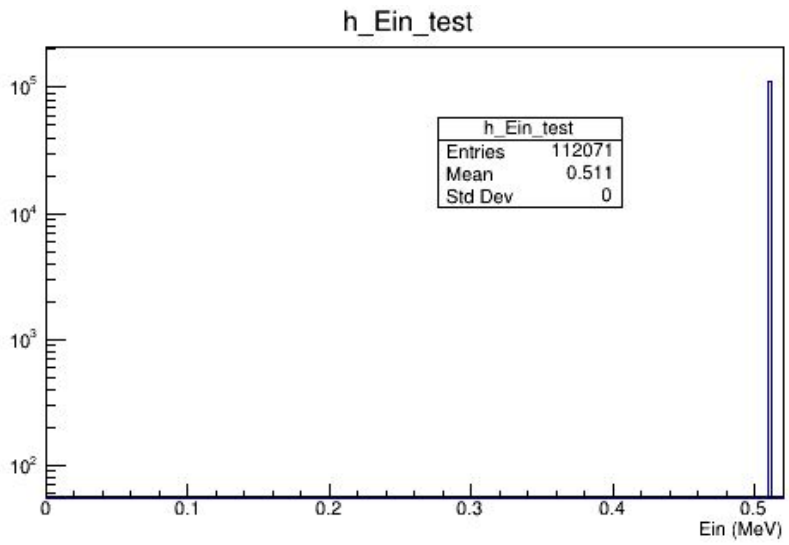






some spread around Z which is not expected.

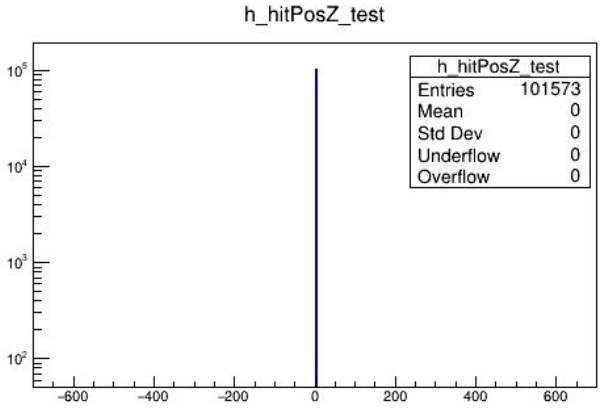
Distribution of Ein



In all the hits the energy of photon before scattering is 511 keV.
This observation suggests that rayleigh scattering might be playing role for the spread around z=0 in the posZ distribution.

Verifying that the spread is due to rayleigh

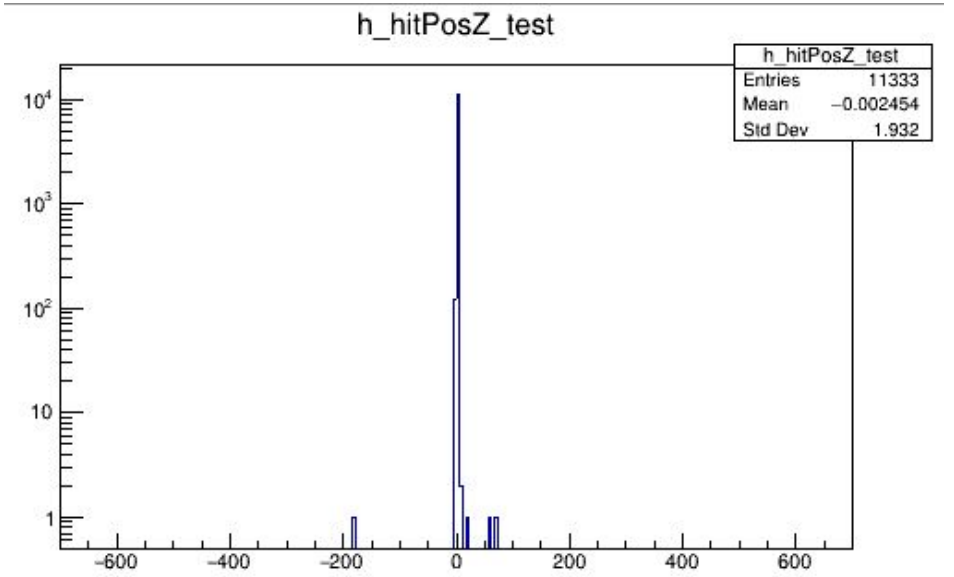
I reject the event if it contains at least one hit from rayleigh scattering.



Now no spread around Z=0 is observed which confirms the reasoning.

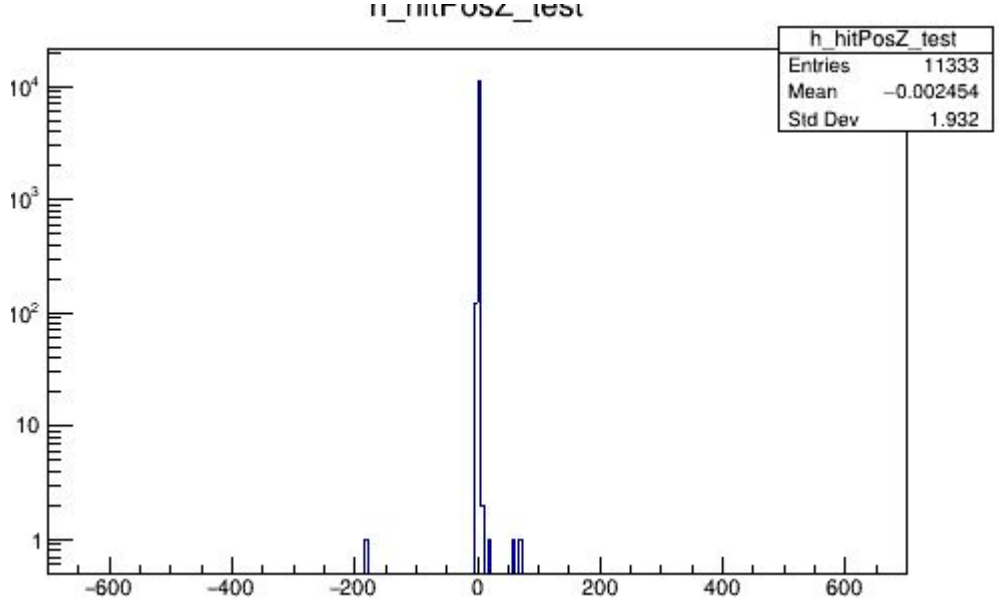
Date: 20/08/25

Result with 10000 events without excluding rayleigh




```
if (proc != "phot") {
  //if(step->GetPreStepPoint()->GetProcessDefinedStep() && step->GetPreStepPoint()->GetProcessDefinedStep()->GetProcessName() != "Rayl"){
  if(proc == "compt" && step->GetPreStepPoint()->GetKineticEnergy() == 0.511) scat_theta = acos(vin.dot(vout));
  //}
```

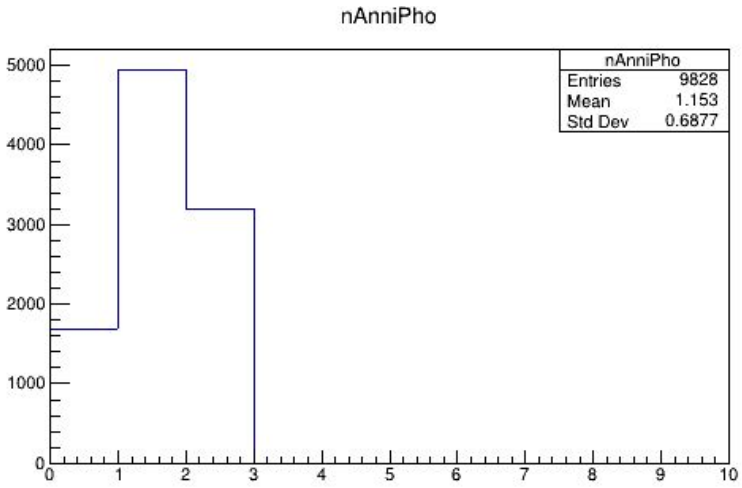
Putting the condition on initial energy in the Analyzer instead of CellSD file at the level of simulation



Things are consistent.

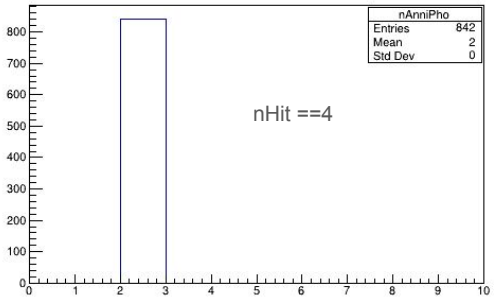
```
float scat_theta_SD = -1000;

if (procName == "Compt" && Ein == 0.511) {scat_theta_SD = HitScatAng_sort[iHit];}
```

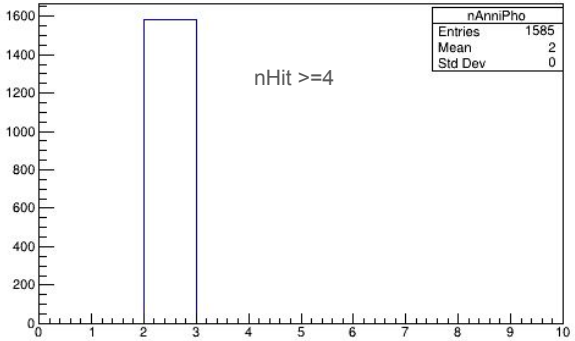
Distribution of Annihilation photons in the event



Filling the histogram in the nhit>0 condition. The 0 annihilation photons in the event corresponding to photoelectric only events.
There are 3192 evts out of 10000 evts where both the annihilation photons does compton scattering.

Looking at the effect of nHit conditions on number



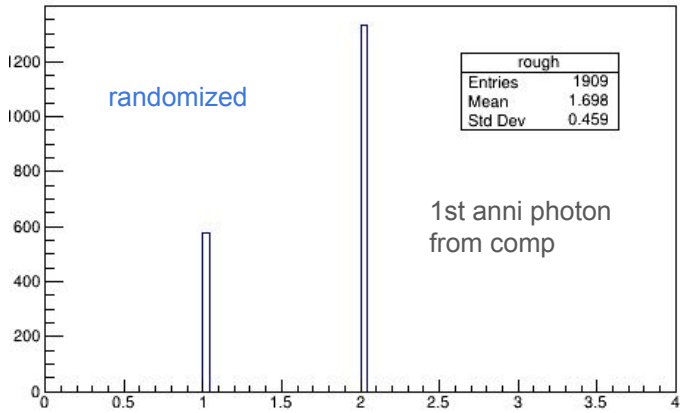


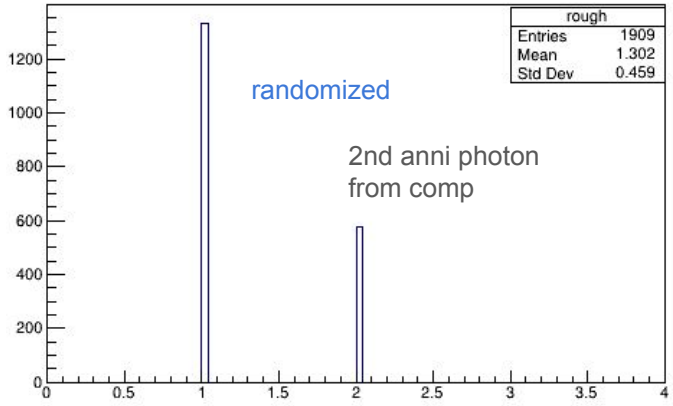

```
if(AnniPho.size()==2 && nHits >=4) h_nAnniPho->Fill(AnniPho.size());
```

I loose too many events after putting the nHit ==4 conditions (from 3192 to 842 only 25% events survive)
While with nHit>=4 condition, around 50% of the events survive.
Here I might be loosing events unnecessarily where hit happens in the same detector tile (large contribution from BGO).

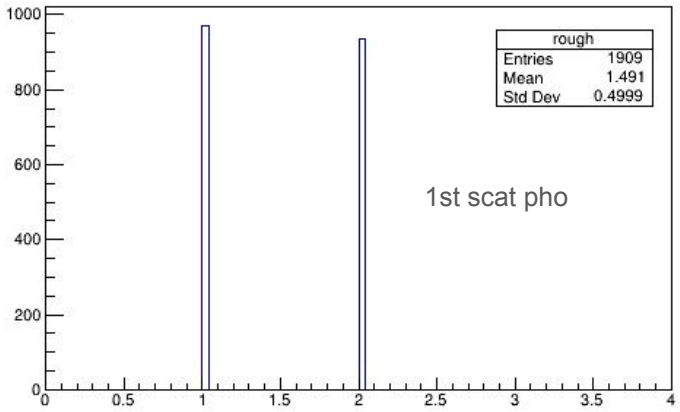
Right now I am going with nHit ==4 condition only just to have the machinery for looking at the timing distribution of 6 combinations.

Distribution of GunIDs of the photons





Don’t understand why the distribution is not flat and why it is inverted for 1st and 2nd


rough
Entries 1909
Mean 1.491
Std Dev 0.4999
1st scat pho
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4

Tagging the gunId

Photon from 1st gun always has trackId=1 parentID=0.

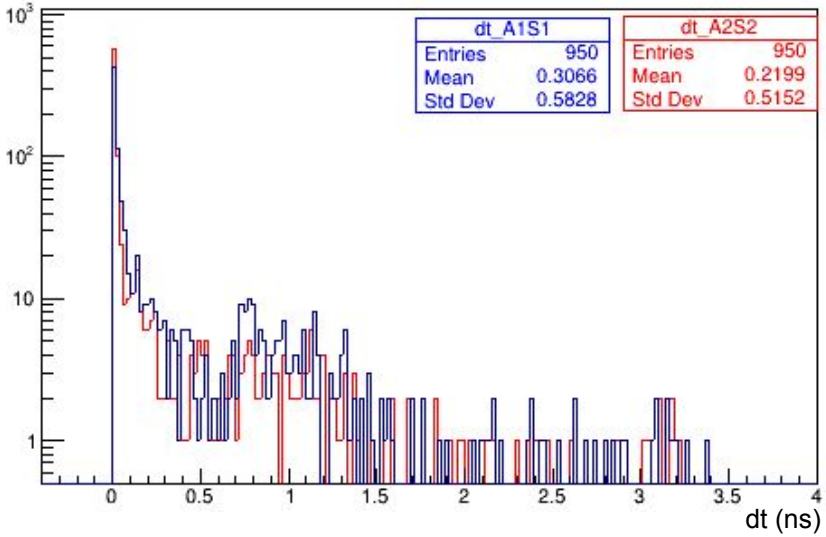
Photon from 2nd gun always has trackId =2 parantID=0.

I am using this information to tag the gun id to the primary photon.

Correcting the time, binning, range in the above plots

Date: 21/08/25

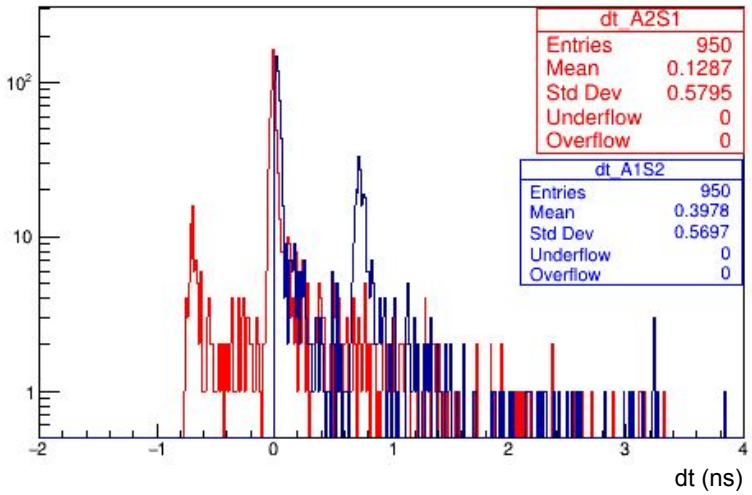
Comparing different distributions after tagging

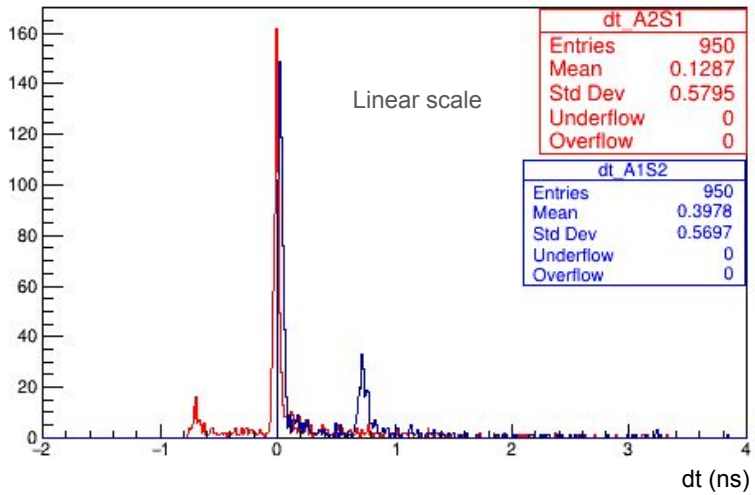


Time difference between actually connected hits

- Event yield seems to be very less 950 out of 100k events (1%).
- Almost similar, which is expected

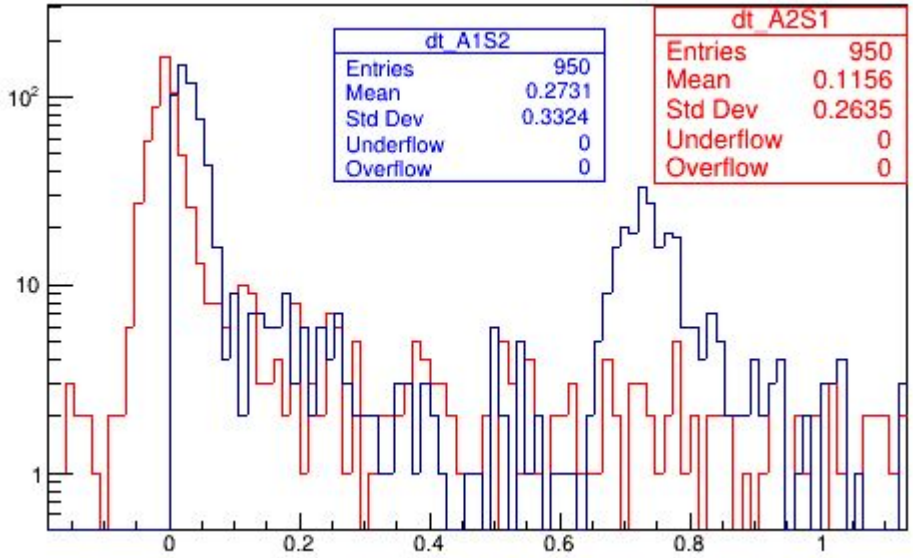
Time difference between actually not connected hits



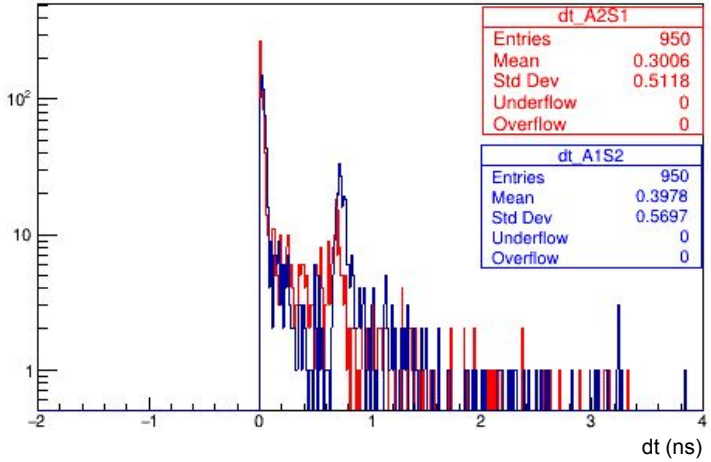


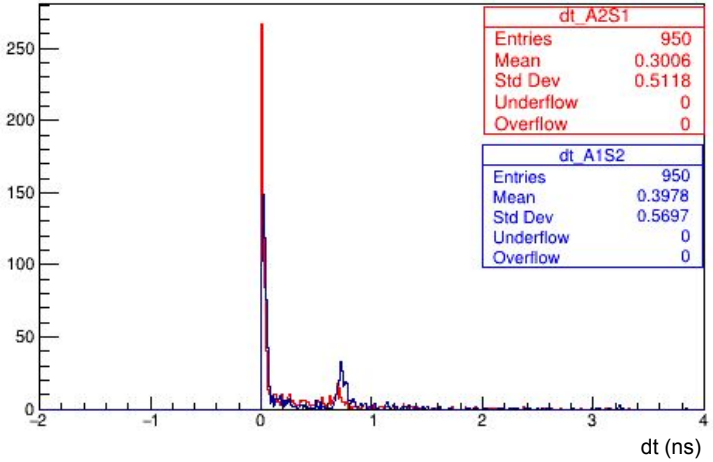
- There is some difference between, but I think they are expected.
- Some in A2S1, some events have dt (t_S1 - t_A2) -ve and +ve because it can happen that after scattered photon from A1 can make a hit before annihilation photon from second gun.
- Looking at the abosolute value of dt in the above case might be helpful.
- Don’t understand the reason for 2nd peak visible in the linear scale

Zooming around the peaks in the previous slide



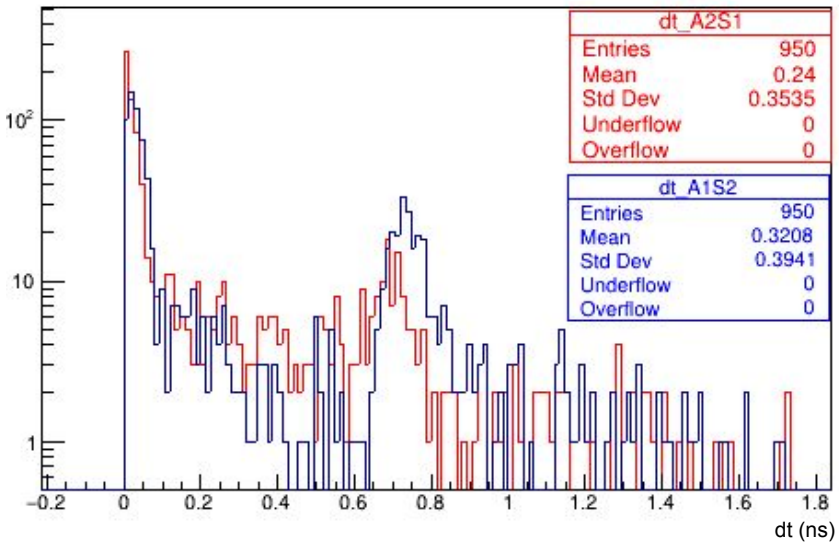
Looking at the absolute value of dt



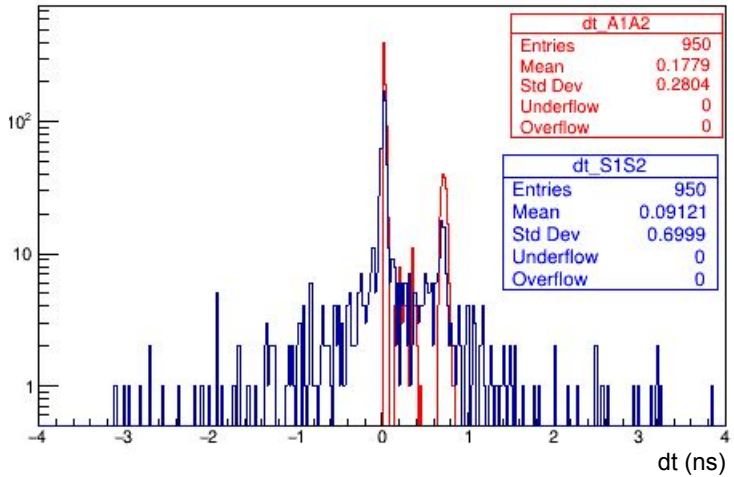


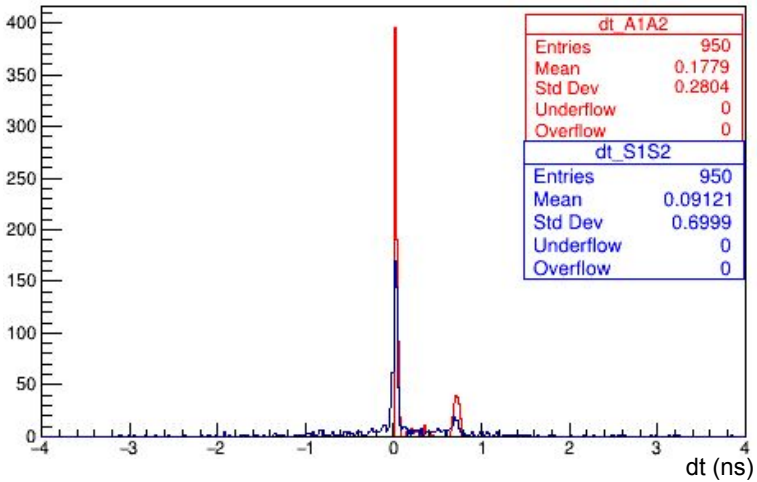
Shape of the distribution seems similar.

Zooming around the peak



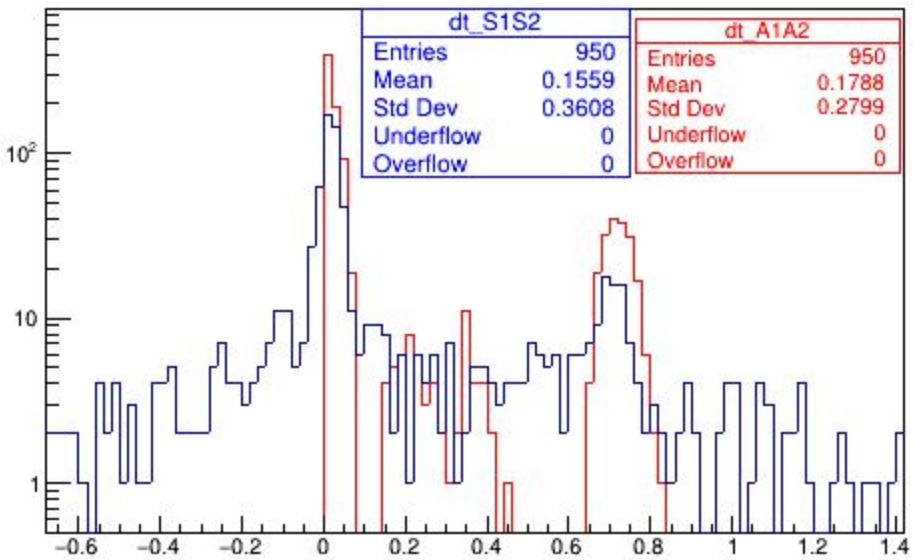
Comparing dt_A1A2 and S1S2



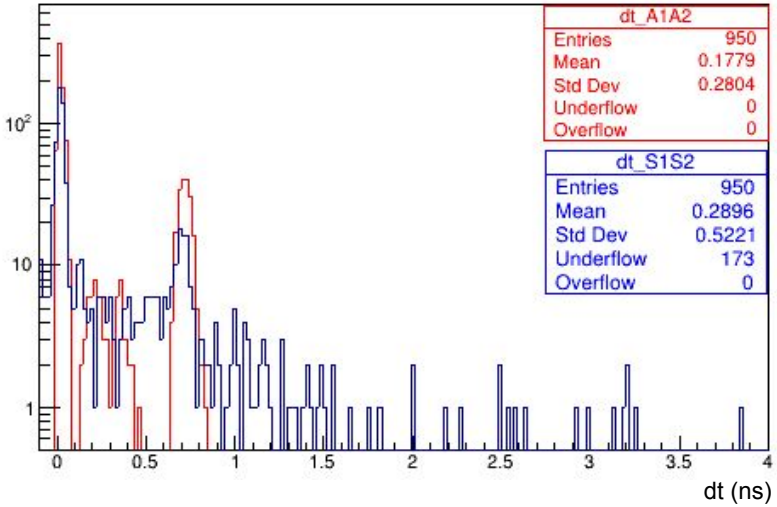


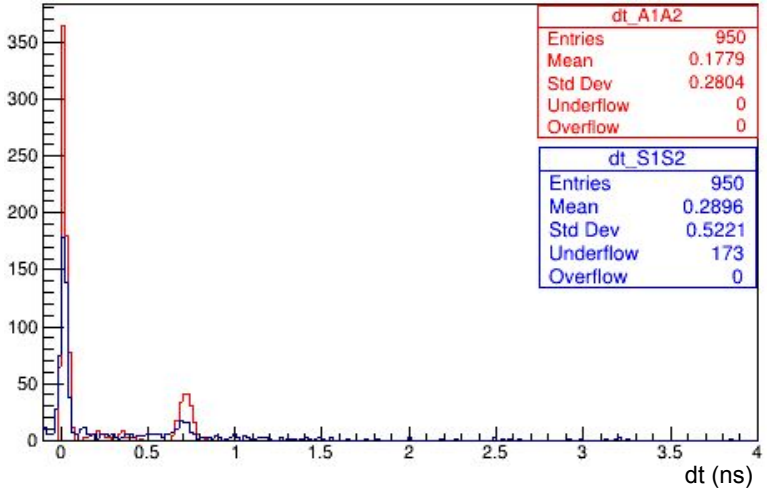
- dt is both negative and positive, because S1 can happen before or after S2. looking at the distribution of absolute value of dt might be helpful.
- Also an additional peak is visible in the linear scale. Which I am not sure why?

Zooming around the peak



Plotting the absolute value of time in S1S2



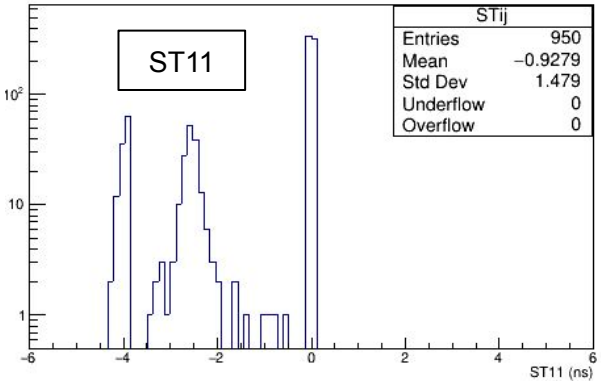


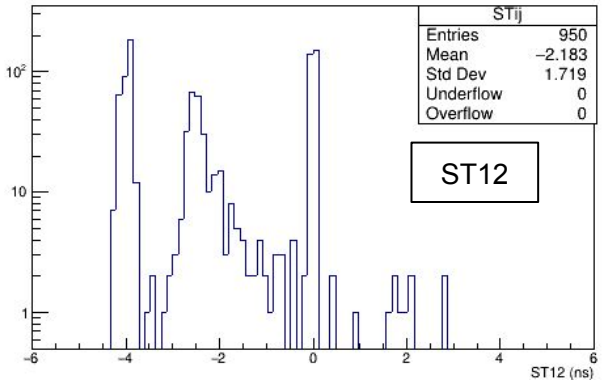
The shape seems to be similar, but height is different.

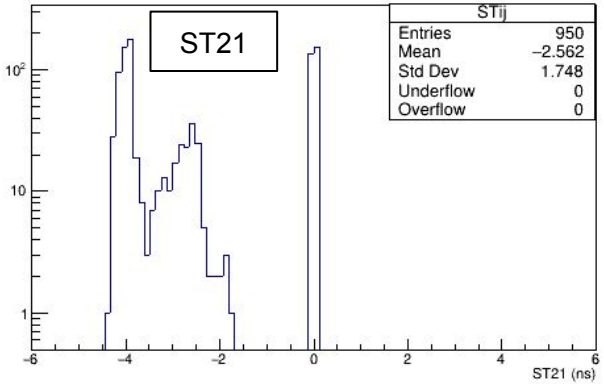
Distributions of STij before connecting the hits

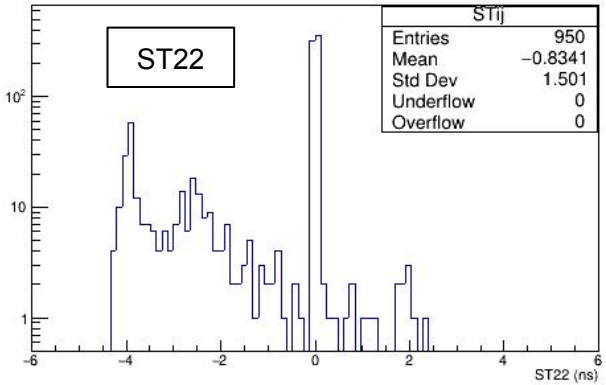
i: index of Annihilation photon hit
j: index of scattered photon hit

- Don't understand why ST11 and ST21 are always -ve

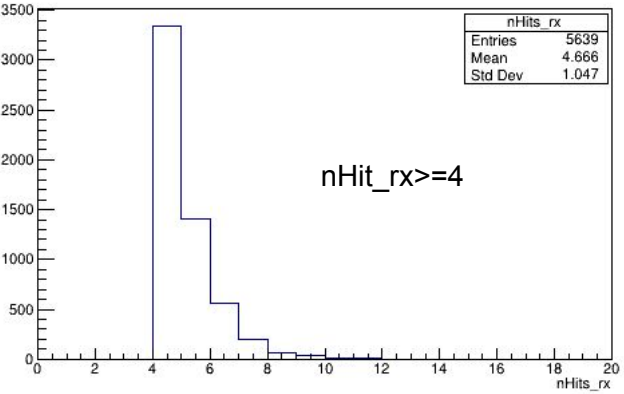


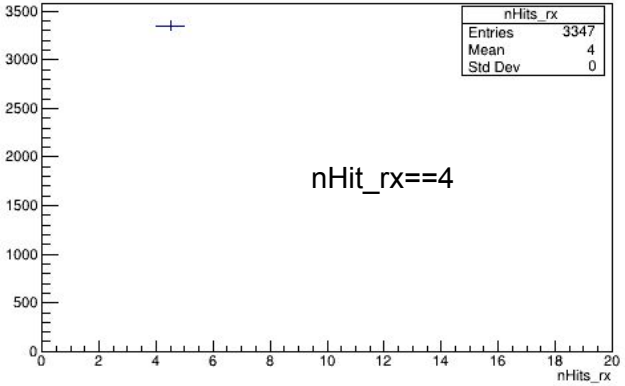


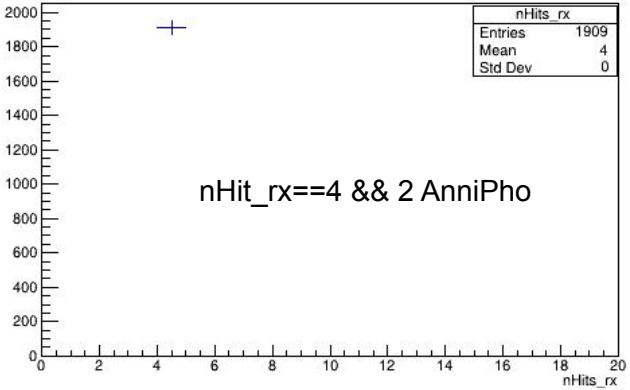


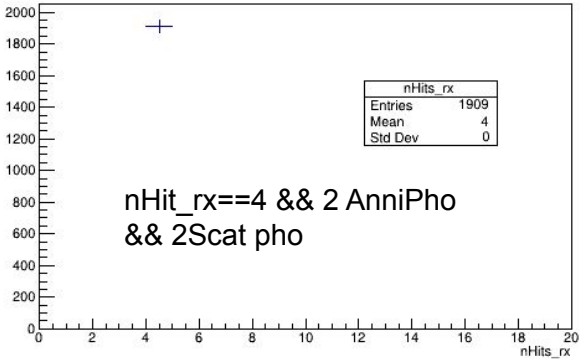


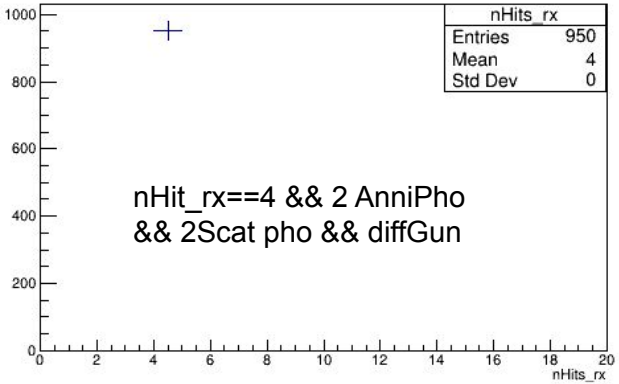
Counting the reduction in events





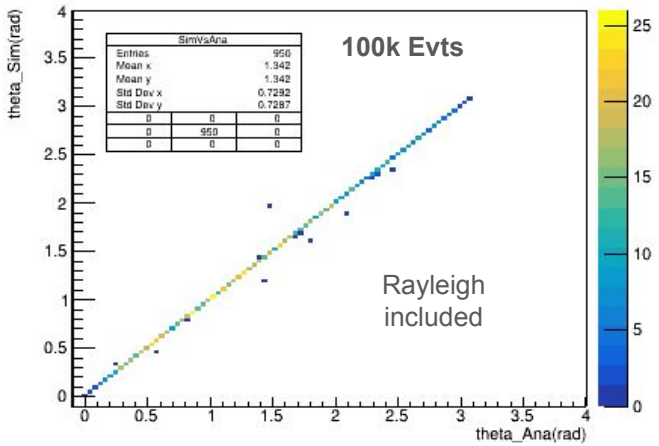


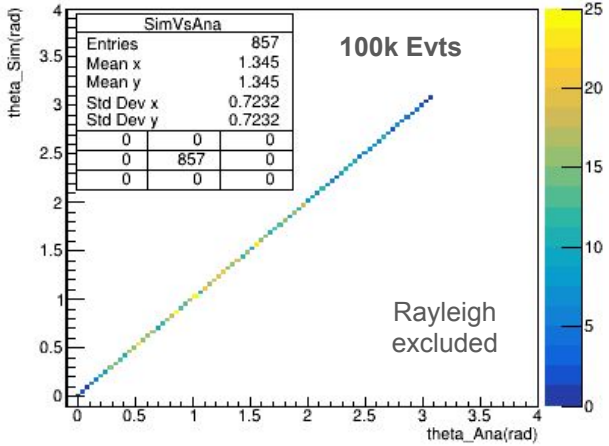




Date: 25/08/25

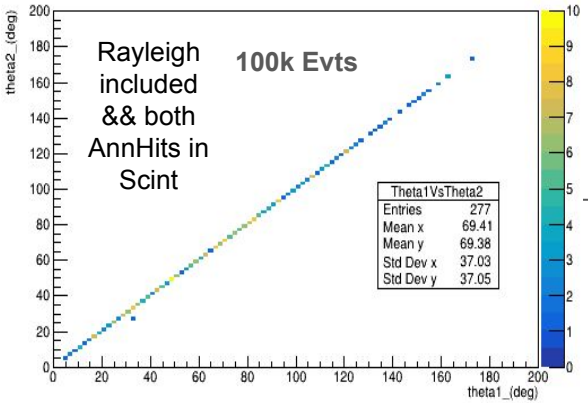
Comparing the Ana vas Sim SD after connecting the hits





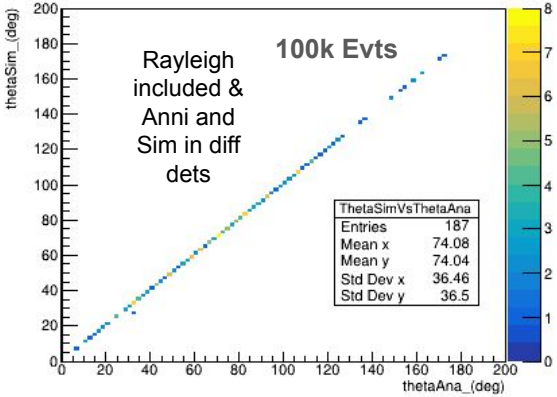
Rayleigh scattering introduces some differences

Testing two different conditions on detMat and detID



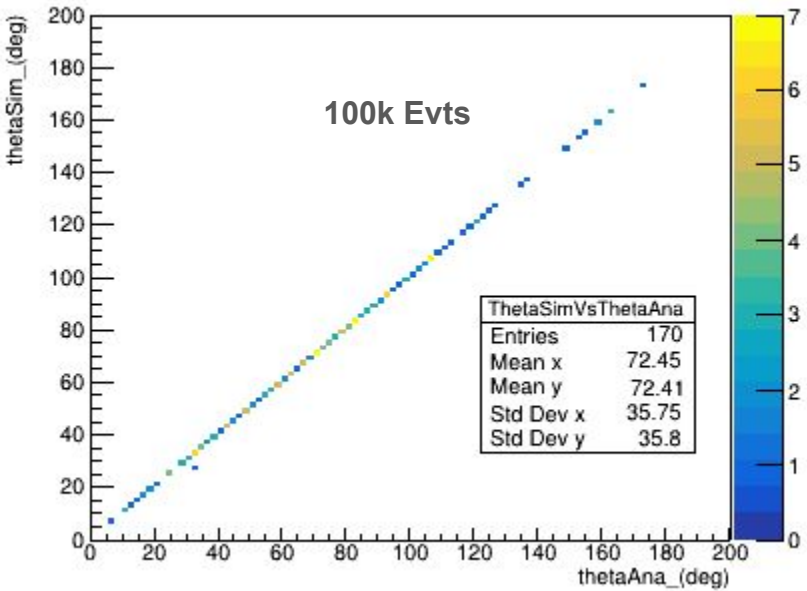

```
if(AnniPho.size()==2 && nHits_RaylEx ==4) {
//if(AnniPho.size()==2) {
  h_nAnniPho->Fill(AnniPho.size());

  if(MatName(AnniPho[0][DetId]) == "Scint" && MatName(AnniPho[1][DetId]) == "Scint"){
```

Loosing too many (~60%) events after asking for both annihilation photon hits in Scintillator

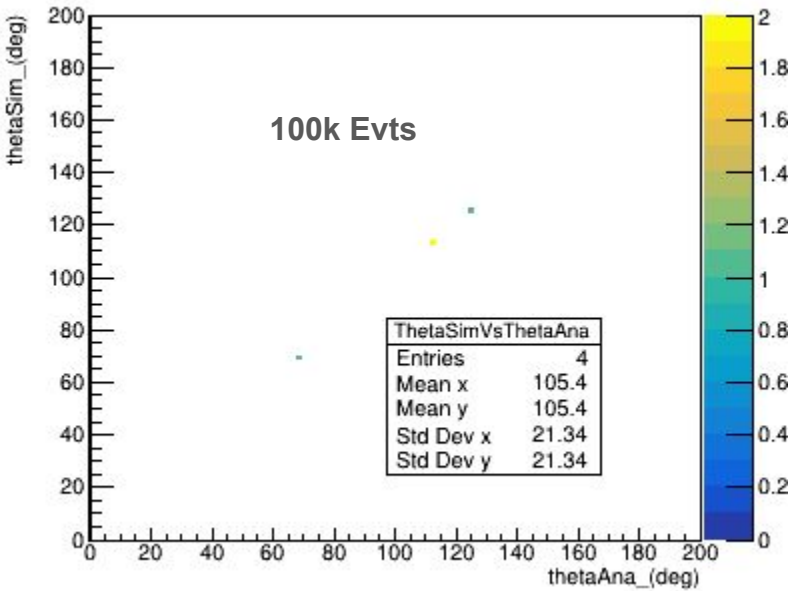



```
    //if (Relv4Hits.size() == 4){
    if (Relv4Hits.size() == 4 &&
        (Relv4Hits[S1][DetId] != Relv4Hits[A1][DetId] && Relv4Hits[S2][DetId] != Relv4Hits[A2][DetI
      float dt_A1S1 = Relv4Hits[S1][time] - Relv4Hits[A1][time];
```

Loosing even more.

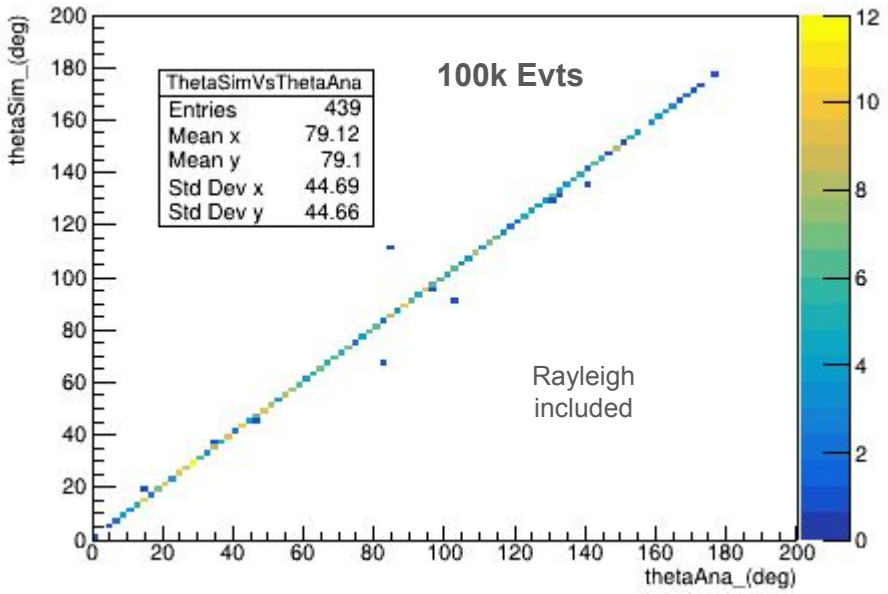


Rayleigh included && both AnnHits in **Scint** also both Anni and Scat pho has different det Id.



Rayleigh included && both AnnHits in **BGO** also both Anni and Scat pho has different det Id.

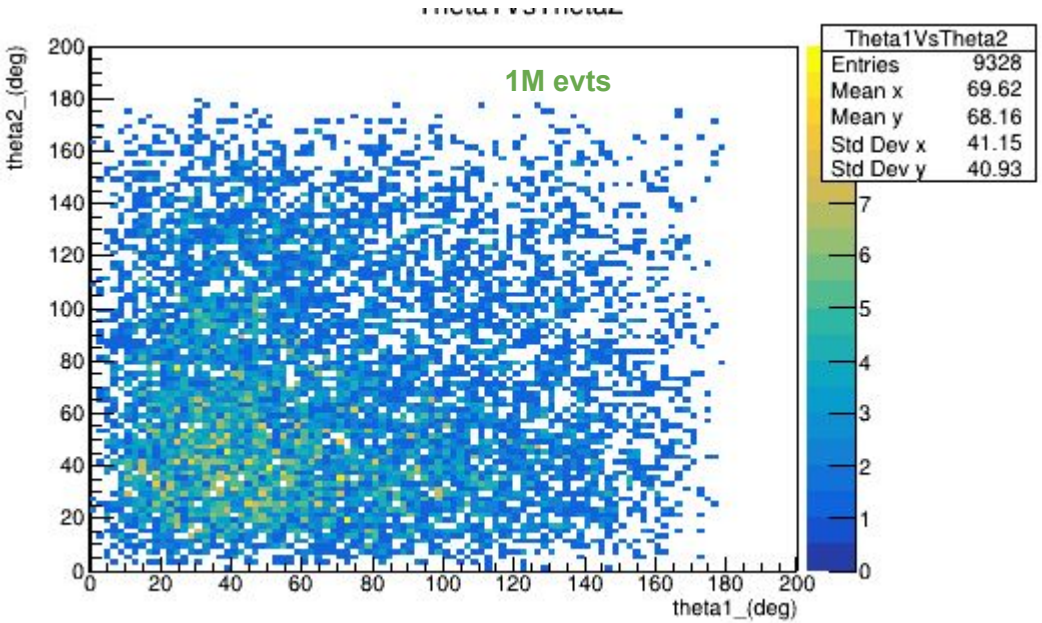
Looking at Annihilation hits in BGO



The entries of this 2d plot shows that out of 950 events, there are 439 events, where either of the annihilation photon is in BGO and scattered photon is in the same tile.

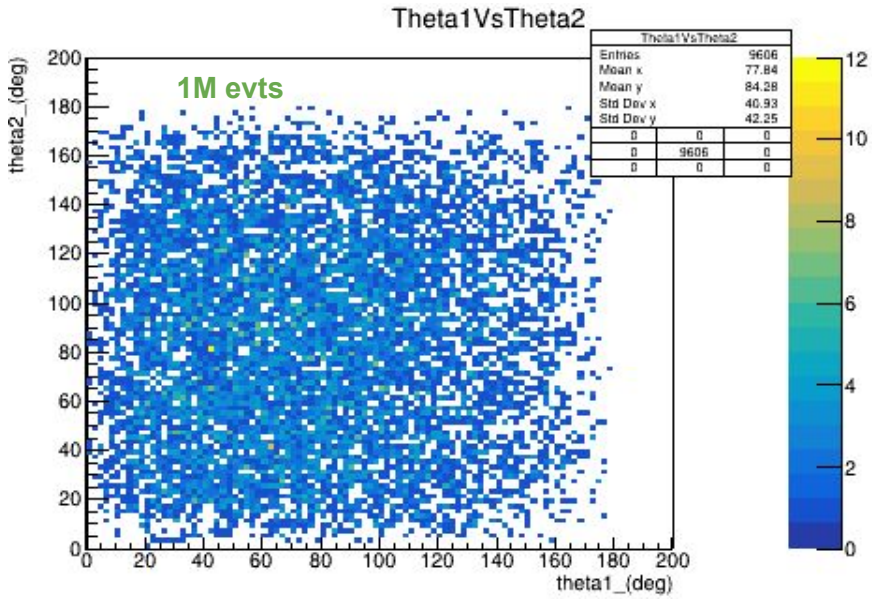
```
if (Relv4Hits.size() == 4){
//if (Relv4Hits.size() == 4 && (Relv4Hits[S1][DetId] == Relv4Hit
  A1S1_sameTile = Relv4Hits[S1][DetId] == Relv4Hits[A1][DetId];
  A2S2_sameTile = Relv4Hits[S2][DetId] == Relv4Hits[A2][DetId];

  if((A1_BGO && A1S1_sameTile) || (A2_BGO && A2S2_sameTile)){
```

Checking the 2D distribution of theta1 and theta 2



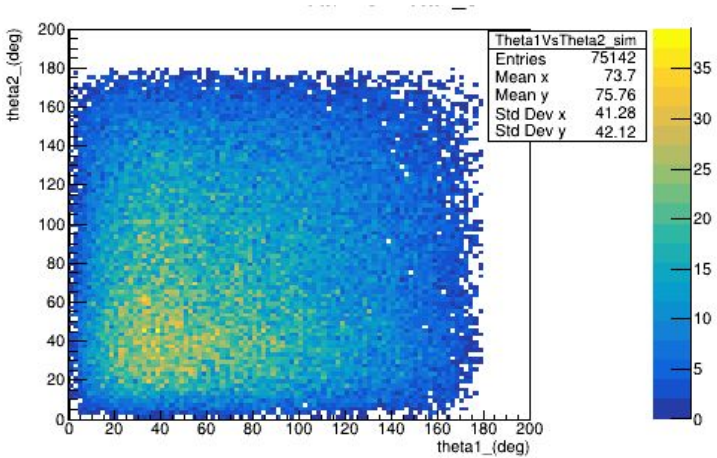
- Taking the angle calculated from simulation to have more event yield to look for.
- Kind of looks like expected one, in the sense that peaked more towards lower scattering angle.

2D distribution for connected hits

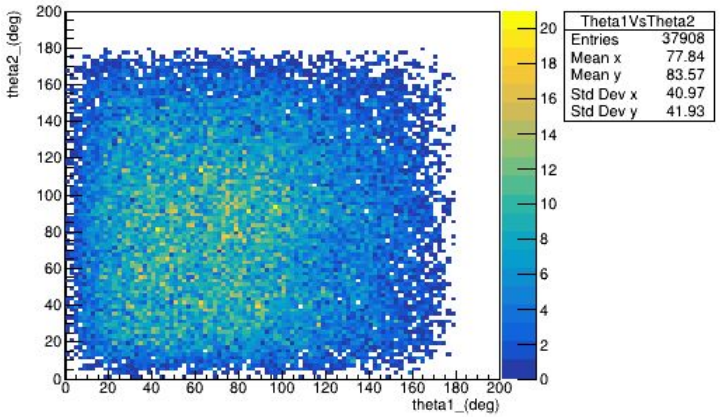


Can’t say much from this.

2D distribution of thetas for 4M Evts



Theta calculated at sim level, registration of scattered photon hit not necessary



Theta calculated from analyzer script from the connected hits, registration of scattered photon hit is necessary.

As mentioned in the JPET paper, lower value of theta is preferred in the 1st case. The second plot seems bit different

2d plots from JPET paper

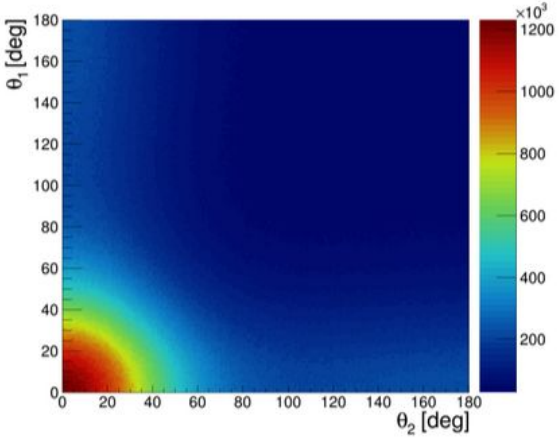


Fig. 11: Event distribution as a function of Compton scatterings angles θ_1 and θ_2 simulated for the case of an ideal detector with 100% registration efficiency. The figure is based on 5.96×10^9 simulated events.

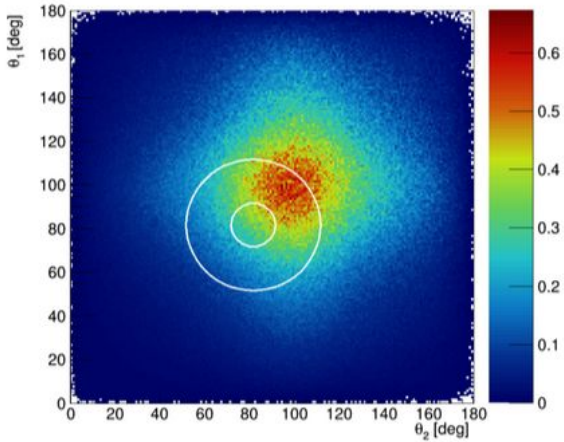
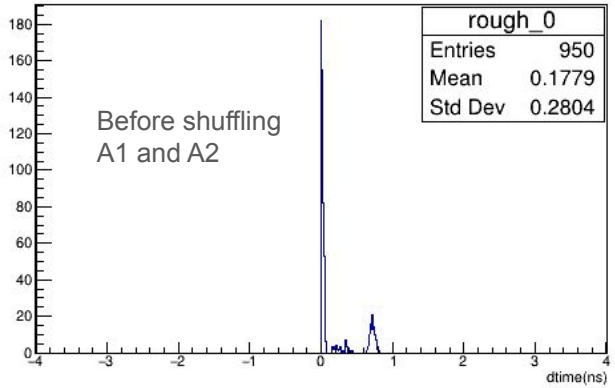
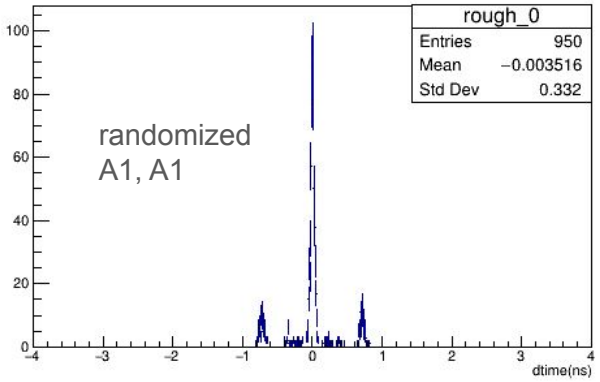


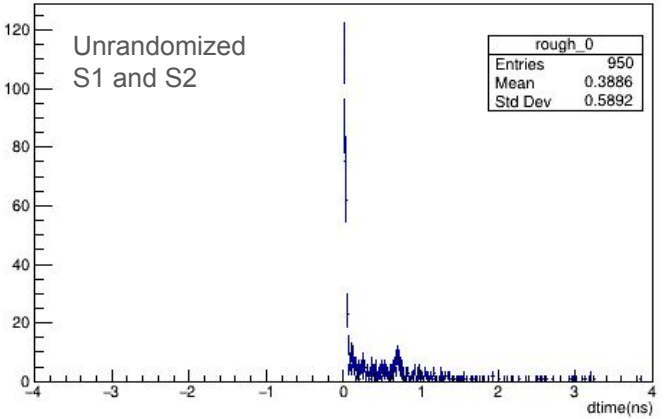
Fig. 12: Idealized (not including event selection criteria) detection efficiency of the J-PET detector as a function of θ_1 vs θ_1. The superimposed circles indicate regions with radius of R = 10° and R = 30° around the point of the highest visibility $\theta_1 = \theta_2 = 81.66°$.

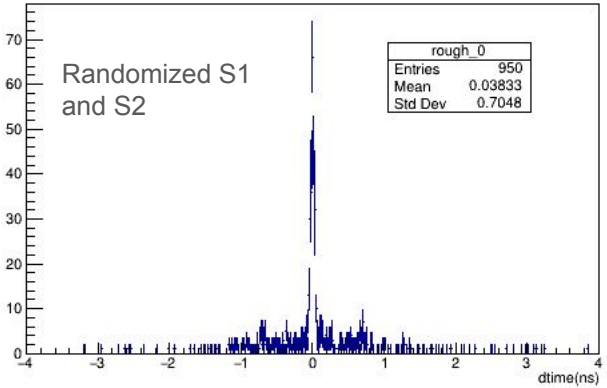
Shuffling A1,A2 and S1,S2

```
std::mt19937 rng(42+jentry);
std::shuffle(AnniPho.begin(), AnniPho.end(), rng);
std::shuffle(ScatPho.begin(), ScatPho.end(), rng);
```



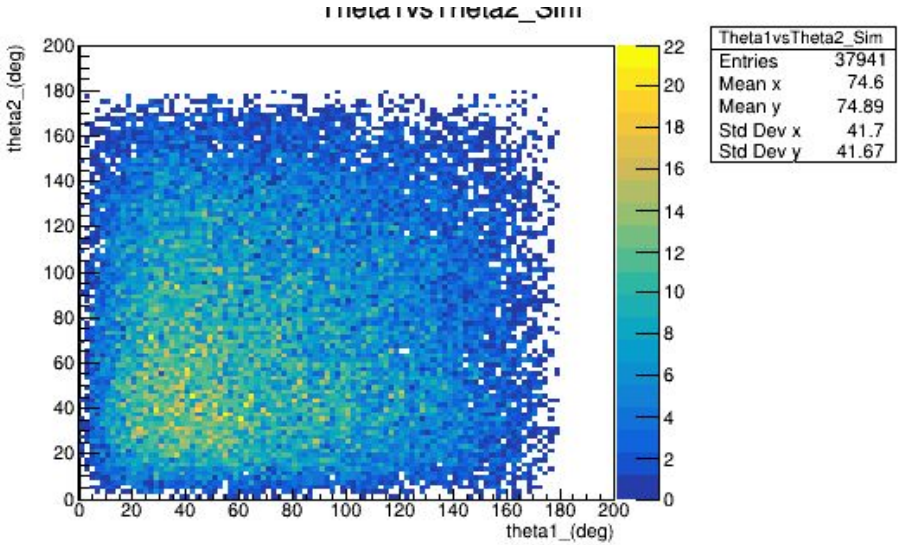


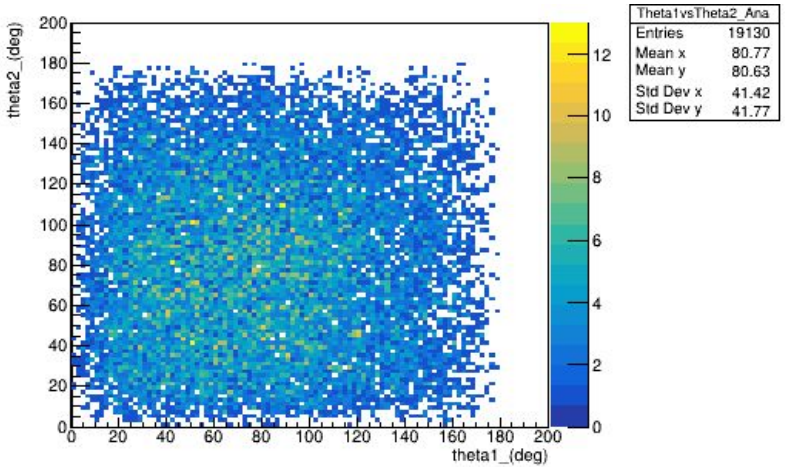




Plots after randomizing A1, A2 and S1, S2

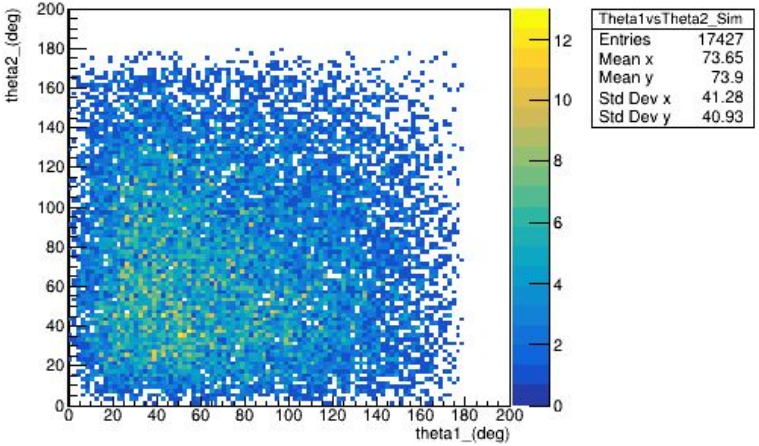
Looking at 2d plots of theta1 and theta2 for 2M evts

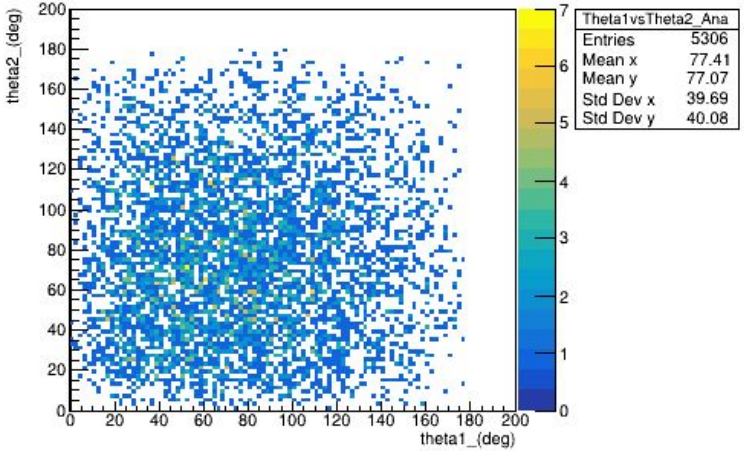


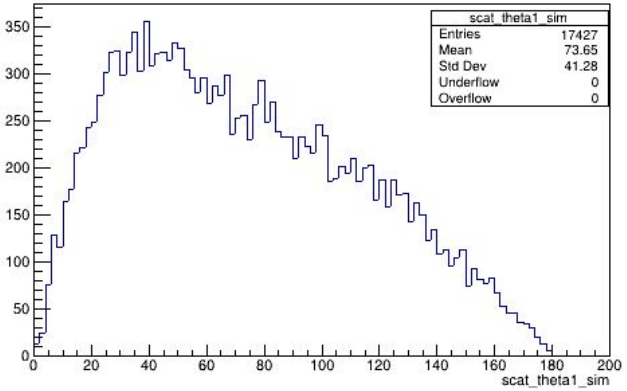


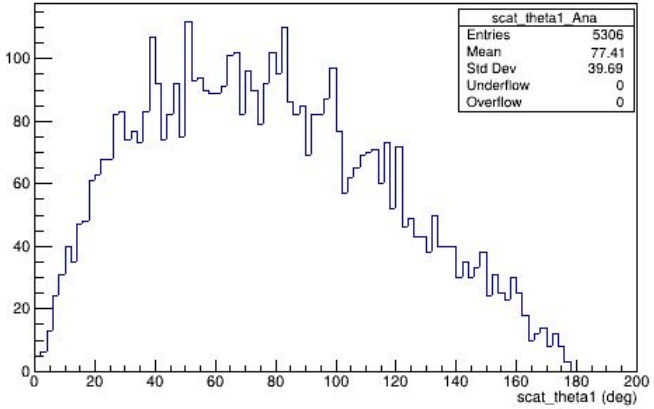
Without putting any condition on Detector ID and detector Mat

When both the hits from annihilation photon are on scint

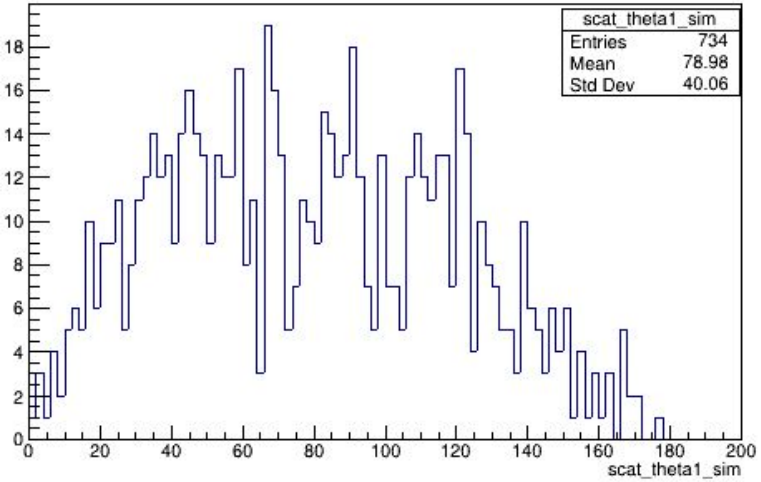


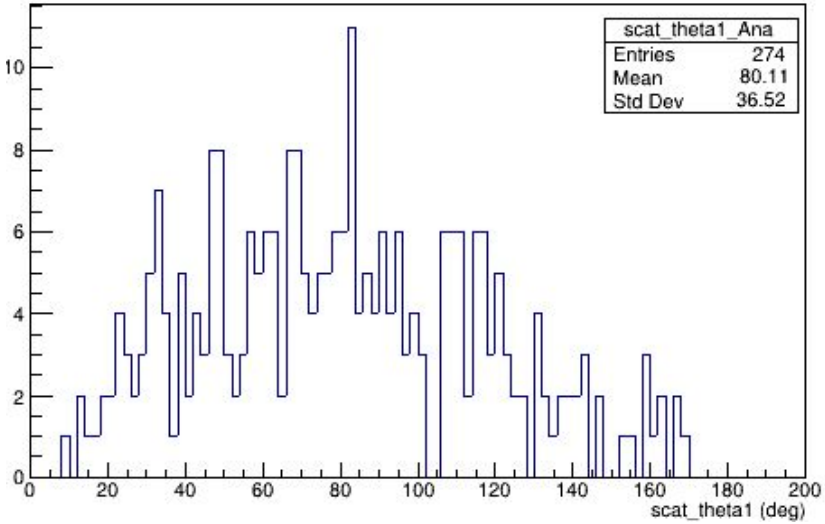






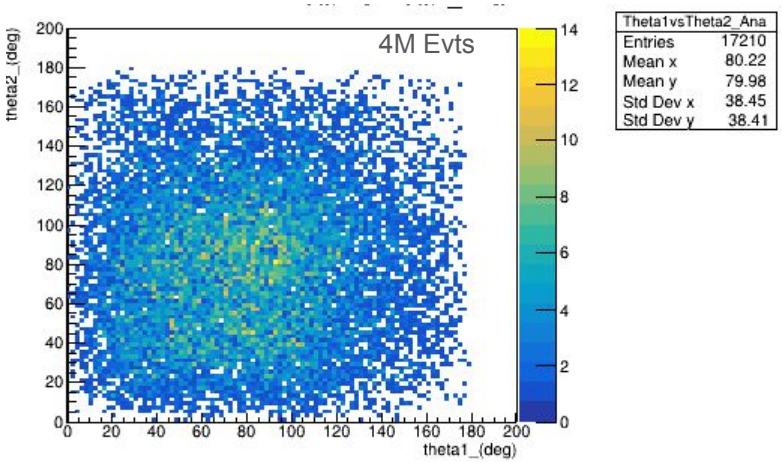
100k evts after changing nHit definition

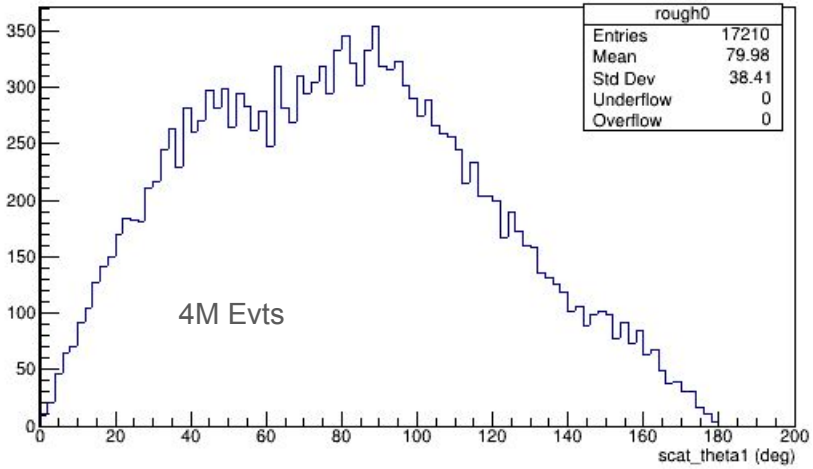




Looks like something worked!! As now scat theta calculated from analysis has clear peak around 82 degs. While sim doesn't.
Earlier no such trend was there. Need to check with 2M events too.

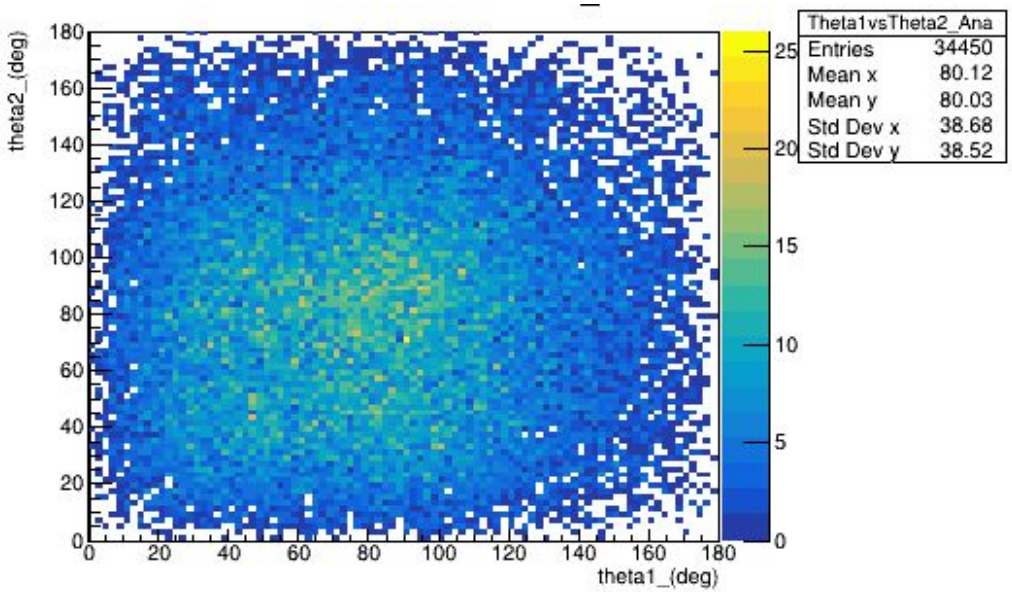
With 4M evts nScatPho>=2

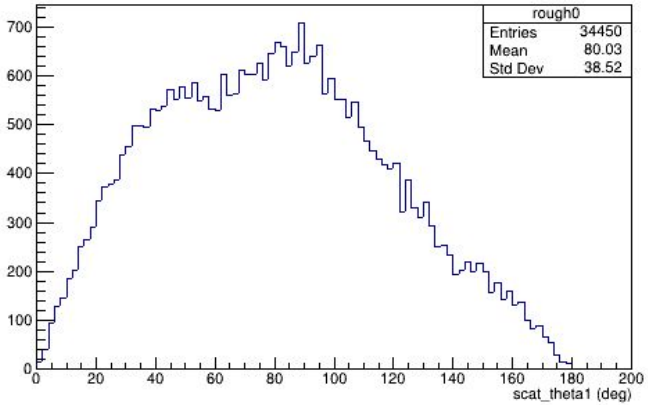




Getting more events above 80!!
Will check with more events.

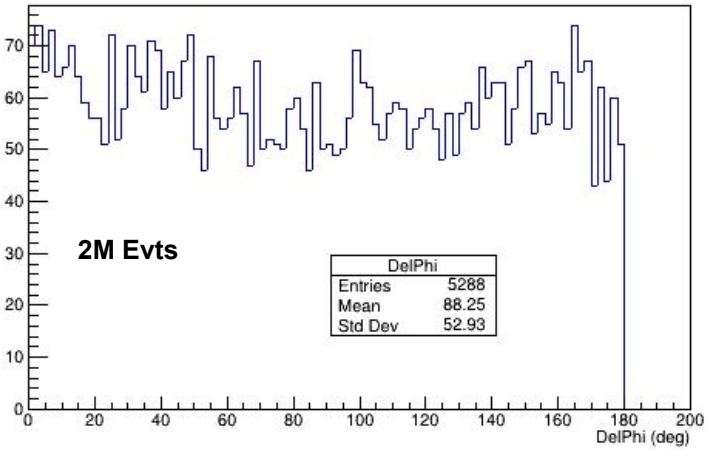
Checking with 8M events



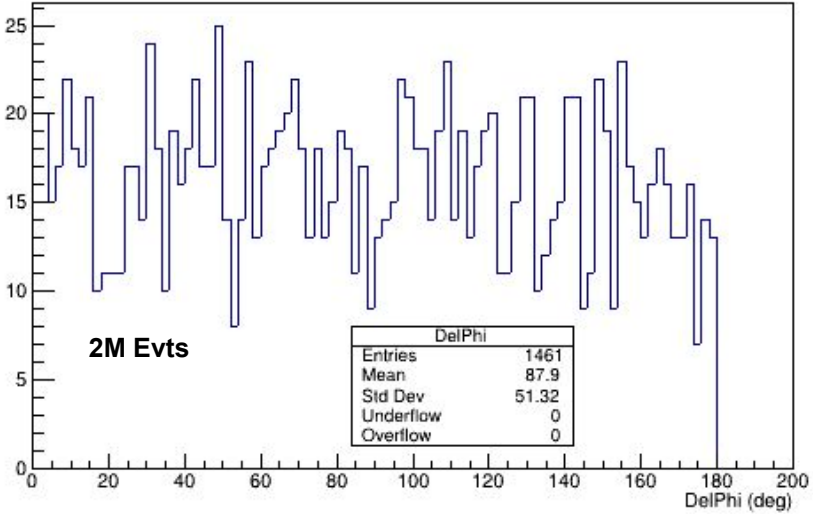


May be doing some normalisation will help.

Distribution of dPhi (Angle between scat planes) nHit ==4

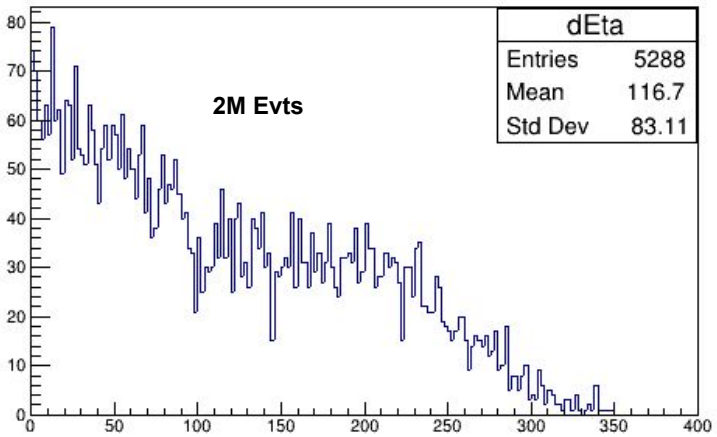


For all theta1, theta2

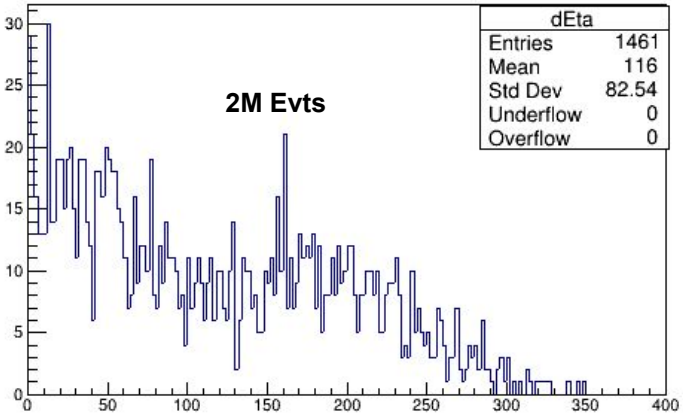


For 50 < theta1<110,
50 < theta2<110

Distribution of dEta nHit ==4

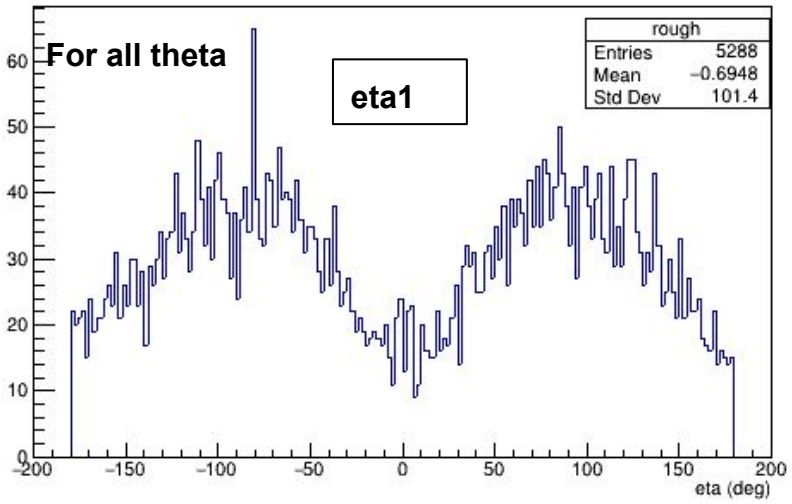


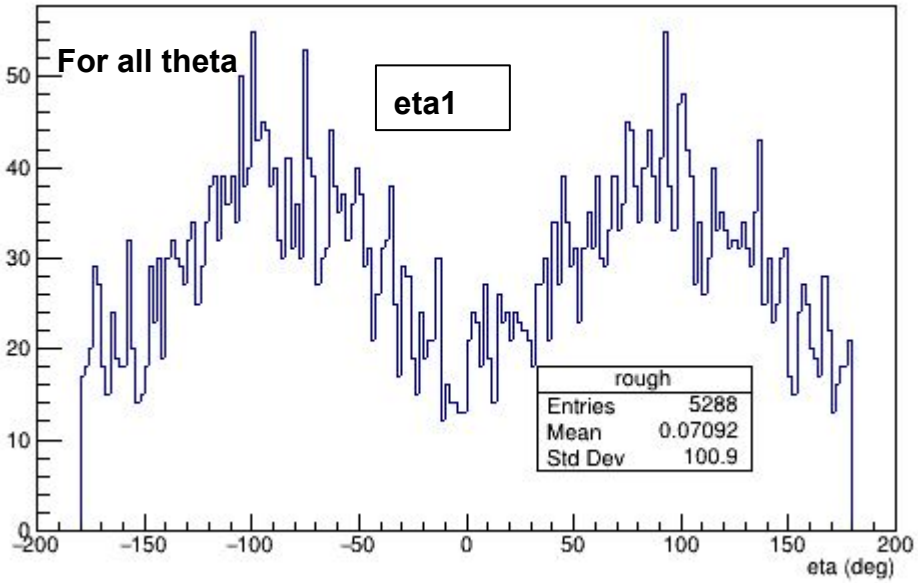
For all theta1, theta2



For 50 < theta1<110,
50 < theta2<110

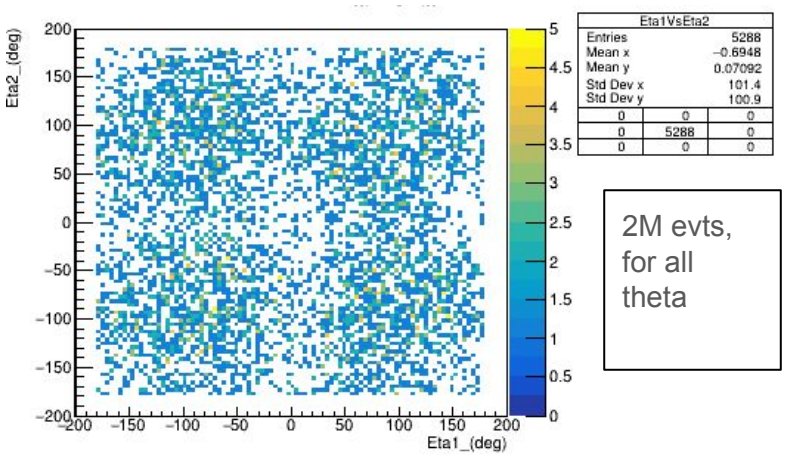
Checkin the 1d hist of the both eta for 2M evts



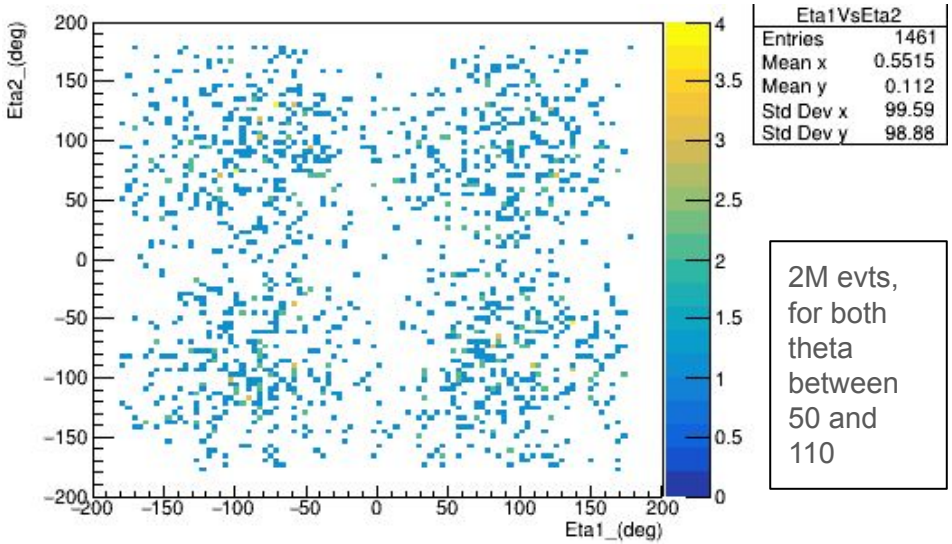


Seems like there is something wrong in the machinery to calculate scattering angle as eta distribution looks like expected one

Checking eta1 vs eta2 for 2M events

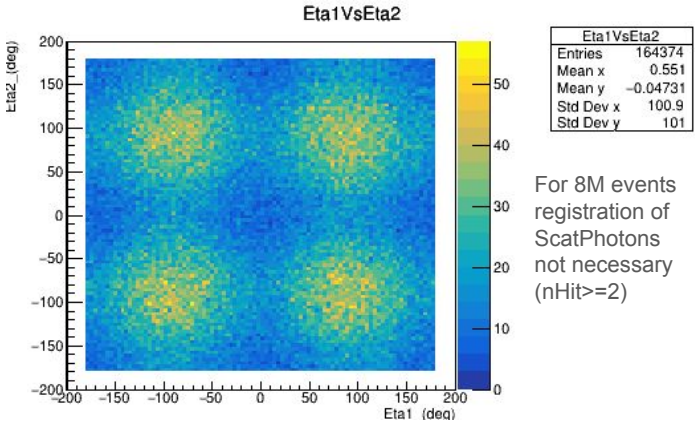


2M evts, for all theta

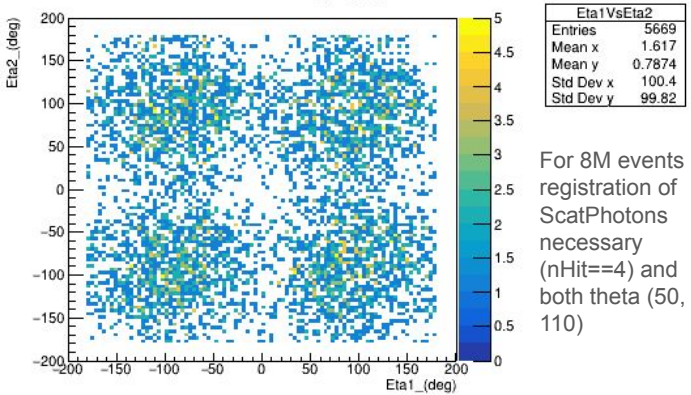


2M evts, for both theta between 50 and 110

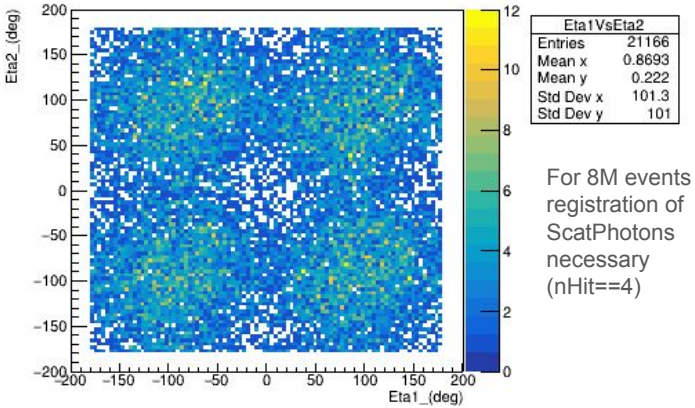
Eta1 vs eta2 for 8M events



For 8M events registration of ScatPhotons not necessary (nHit>=2)

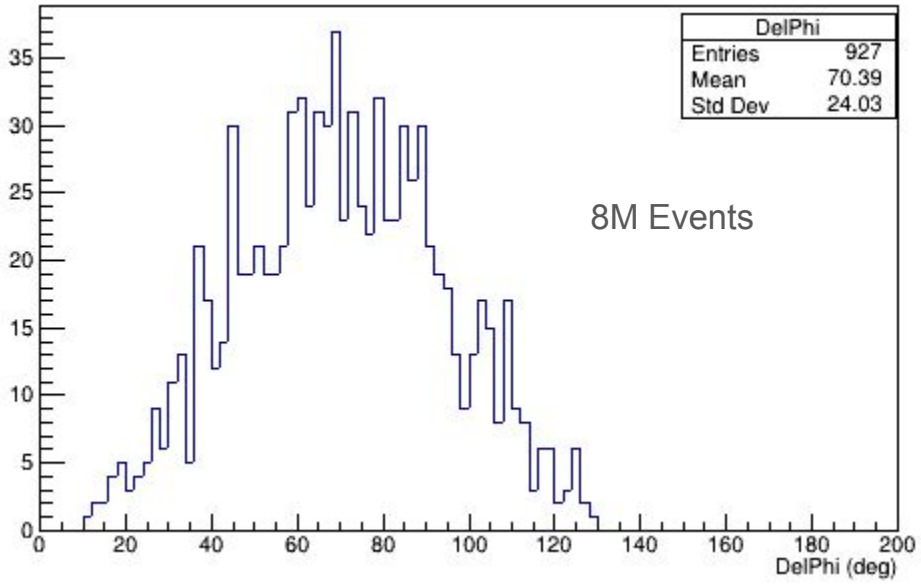


For 8M events registration of ScatPhotons necessary (nHit==4) and both theta (50, 110)

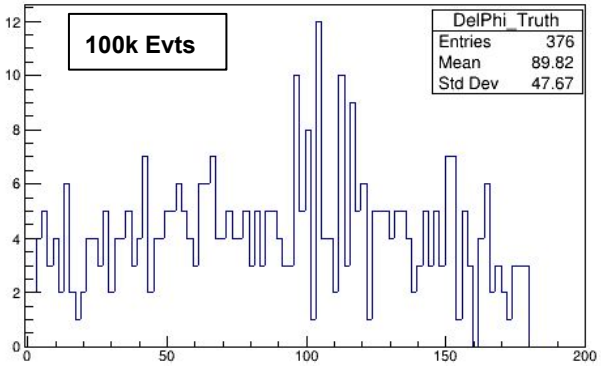


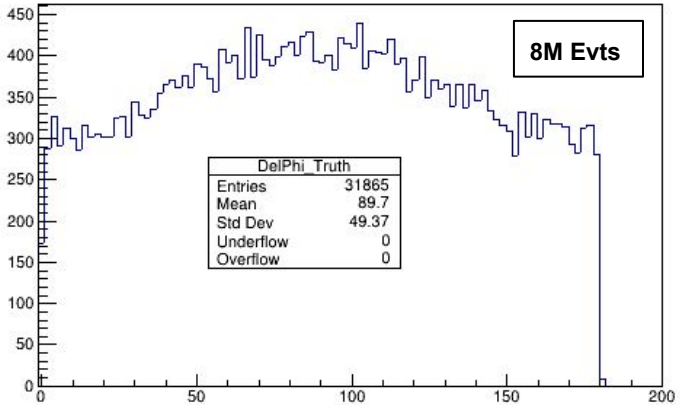
For 8M events registration of ScatPhotons necessary (nHit==4)

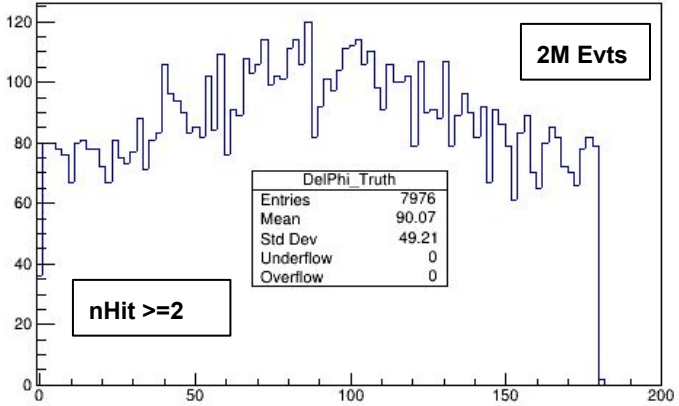
When directly apply condition that both eta (80,110)



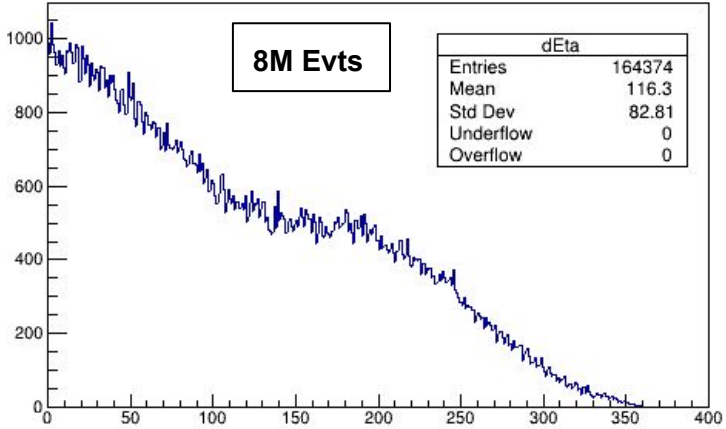
Delphi Using truth level information



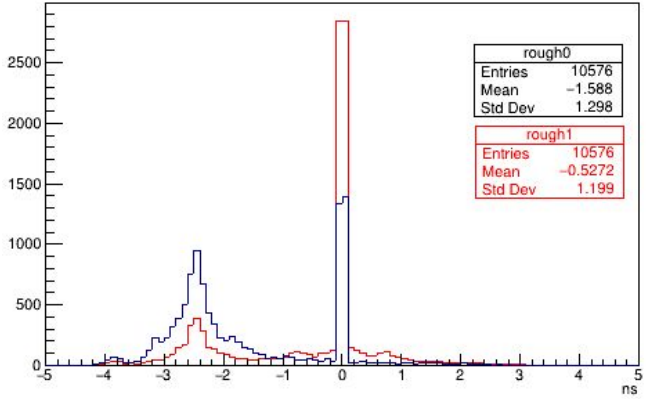




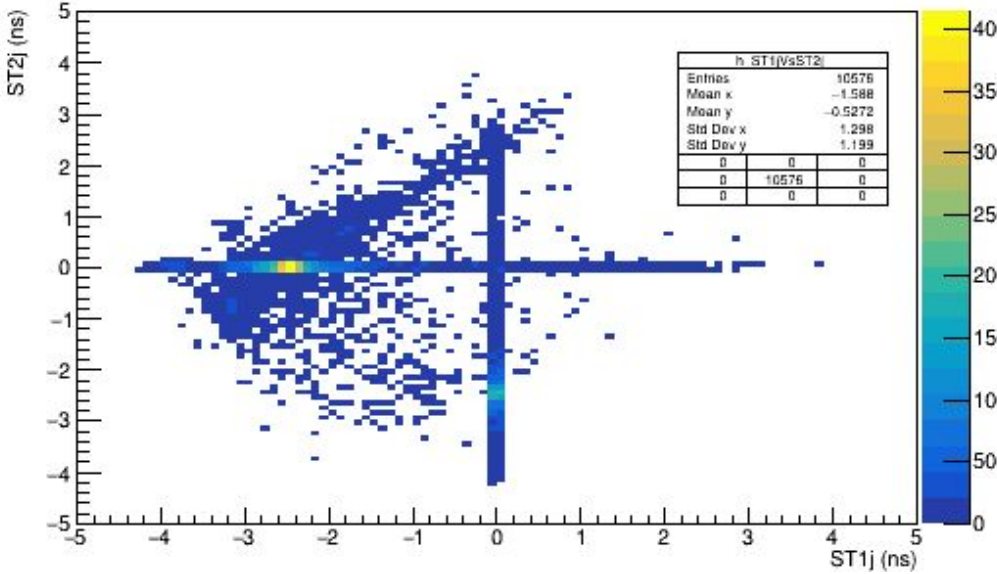
Distribution of dEta for 8M events



For 2M events

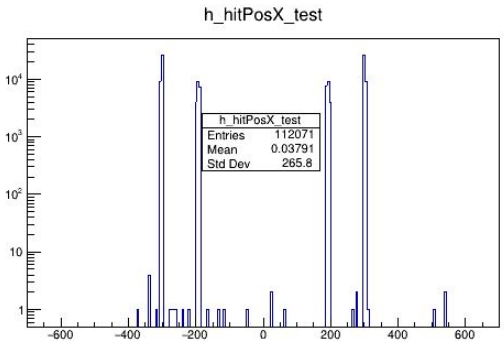


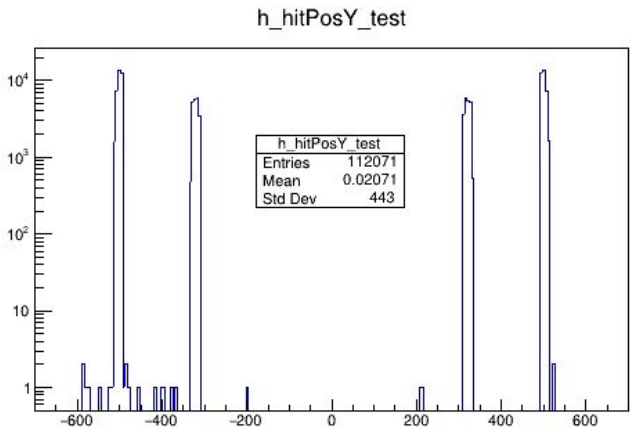
Blue: ST1j
Red: ST2j

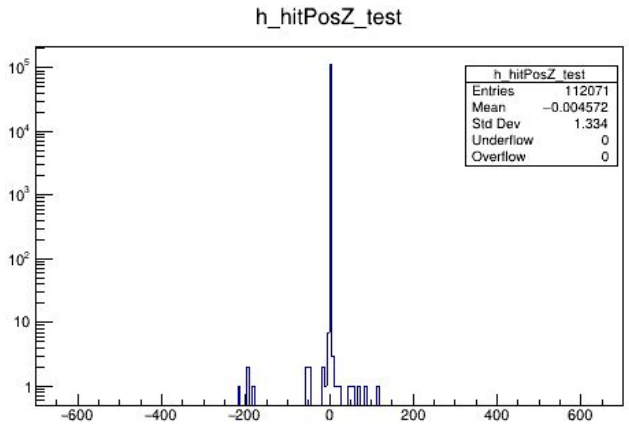


Back Up

Excluding rayleigh of the prestep point








```
if (proc != "phot") {
  if(step->GetPreStepPoint()->GetProcessDefinedStep() && step->GetPreStepPoint()->GetProcessDefinedStep()->GetProcessName() != "Rayl"){
    if(proc == "compt" && step->GetPreStepPoint()->GetKineticEnergy() == 0.511) scat_theta = acos(vin.dot(vout));
    }
```

The spread in the z coordinate doesn’t make sense here as direction of the incident photon is fixed in the xy plane.

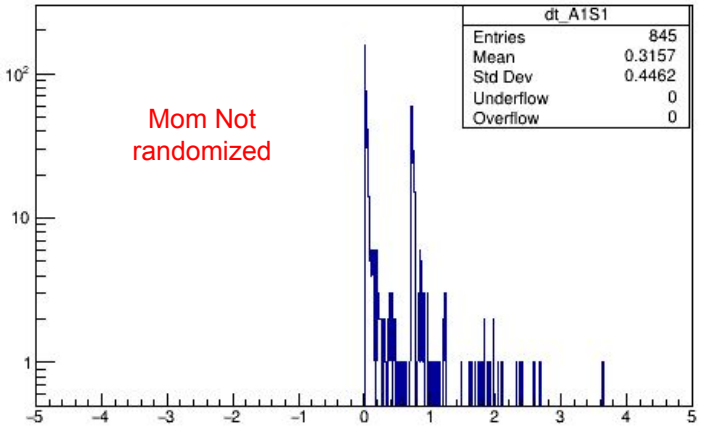
Reason for rayleigh scattering condition not working

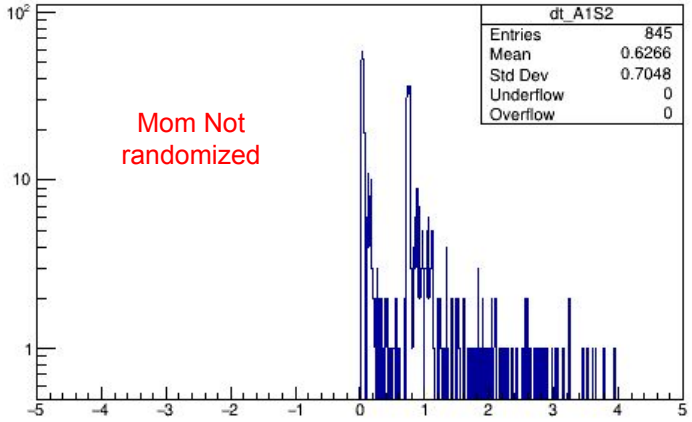
```
*******************************************************************************************************
* G4Track Information:   Particle = gamma,   Track ID = 1,   Parent ID = 0
*******************************************************************************************************

Step#    X(mm)    Y(mm)    Z(mm) KinE(MeV)  dE(MeV) StepLeng TrackLeng  NextVolume ProcName
    0        0        0        0     0.511        0        0         0       World initStep
    1      186      310        0     0.511        0      362       362     InStrip Transportation
    2      199      331        0     0.511        0       24       386       World Transportation
    3      297      495        0     0.511        0      192       578 OuterMostStrip Transportation
    4      299      499        0     0.511        0     4.58       582 OuterMostStrip Rayl
    5      304      494     1.28     0.511        0     6.89       589       World Transportation
    6      539      214     70.5     0.511        0      373       962 OuterMostStrip Transportation
    7      540      213     70.6     0.266 0.000537    0.423       962 OuterMostStrip compt
    8      539      213     70.9         0   0.0905    0.538       963 OuterMostStrip phot
```

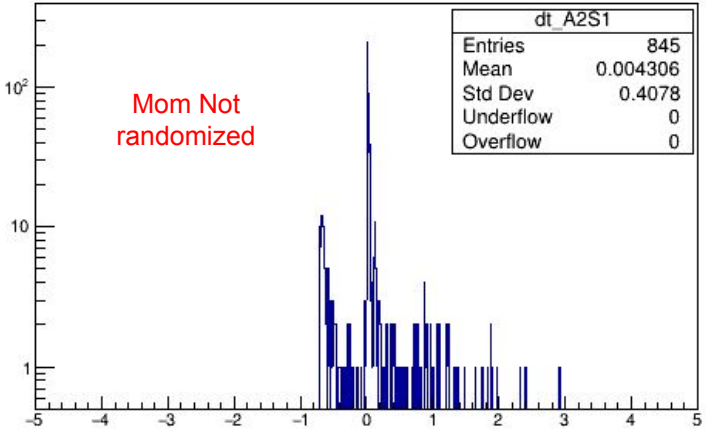
```
f (proc != "phot") {
 if(step->GetPreStepPoint()->GetProcessDefinedStep() && step->GetPreStepPoint()->GetProcessDefinedStep()->GetProcessName() != "Rayl"){
   if(proc == "compt" && step->GetPreStepPoint()->GetKineticEnergy() == 0.511) scat_theta = acos(vin.dot(vout));
   }
```

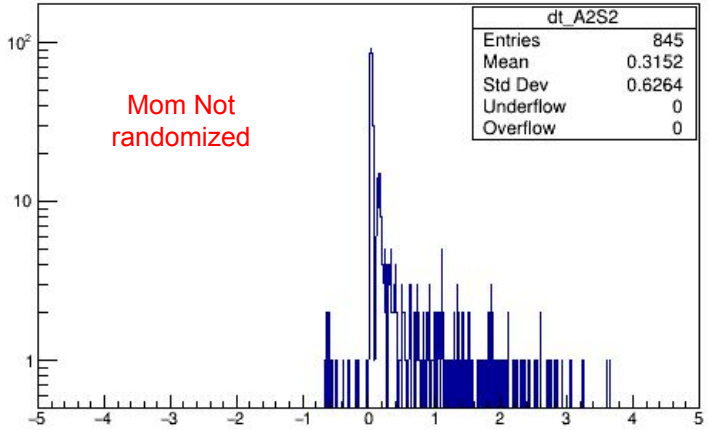
The process at prestep point at compton in transportation. That's why the condition was not working. Here it seems like I can make use of the Tracklength information. But for time being now, I am simply rejecting the events where at least one rayleigh is present. By doing that I lose around 10% of the events.

Very crude version of delta time distribution of 6 combinations

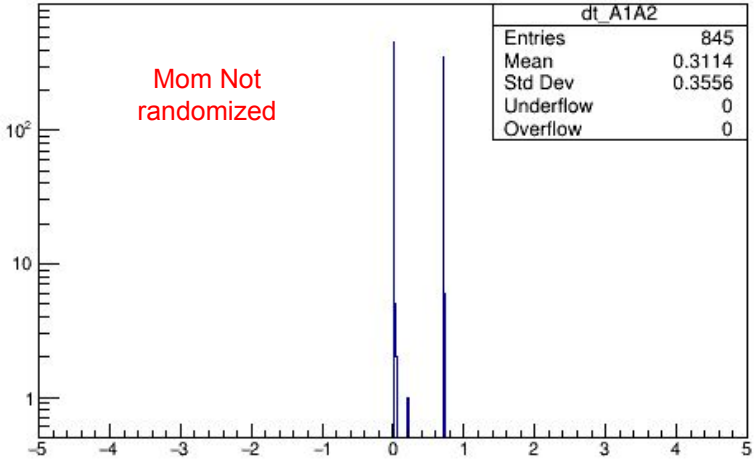


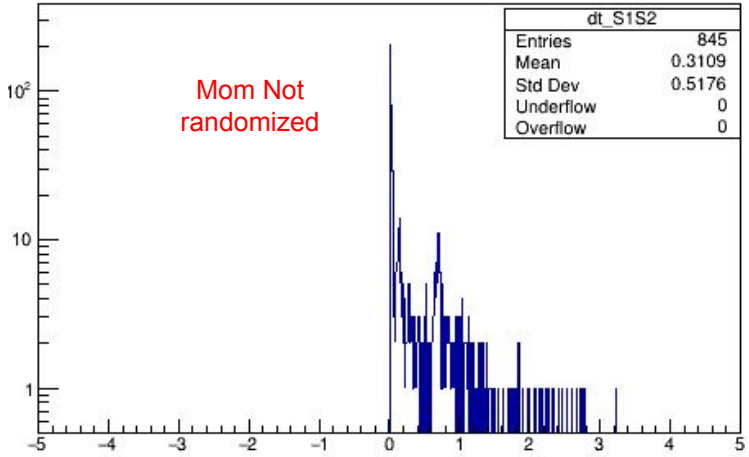




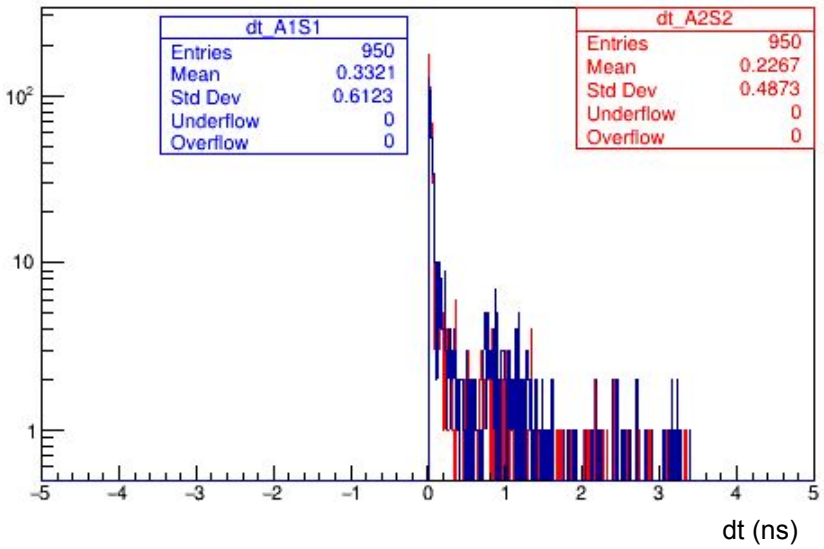


contd..


dt_A1A2
Entries 845
Mean 0.3114
Std Dev 0.3556
Underflow 0
Overflow 0
Mom Not randomized
10^2
10
1
-5 -4 -3 -2 -1 0 1 2 3 4 5


dt_S1S2
Entries 845
Mean 0.3109
Std Dev 0.5176
Underflow 0
Overflow 0
Mom Not randomized
10^2
10
1
-5 -4 -3 -2 -1 0 1 2 3 4 5

Comparing different distributions after tagging



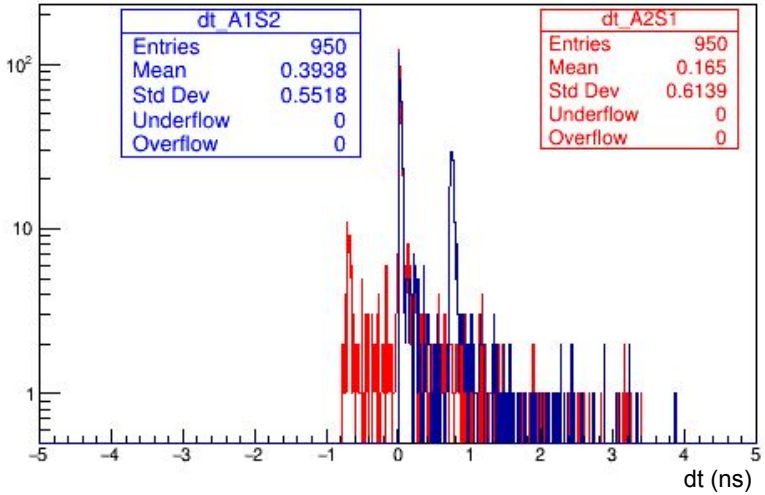
Time difference between actually connected hits

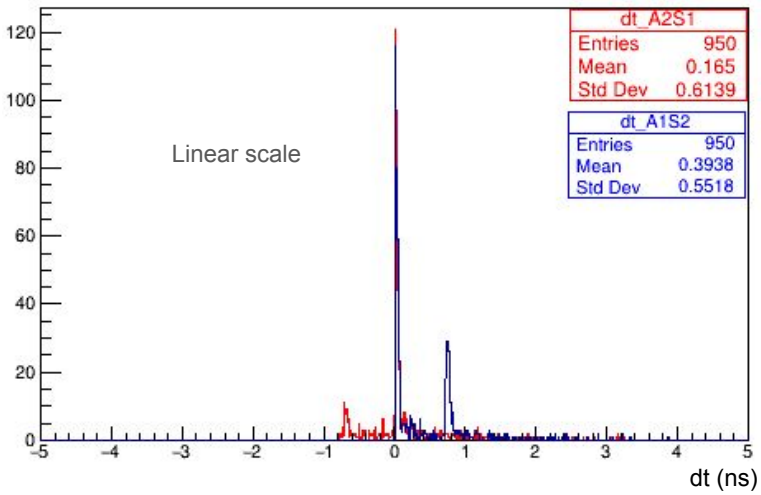
Don't use these timing plots in the final slides

- Event yield seems to be very less 950 out of 100k events (1%).
- Almost similar, which is expected

Date: 21/08/25

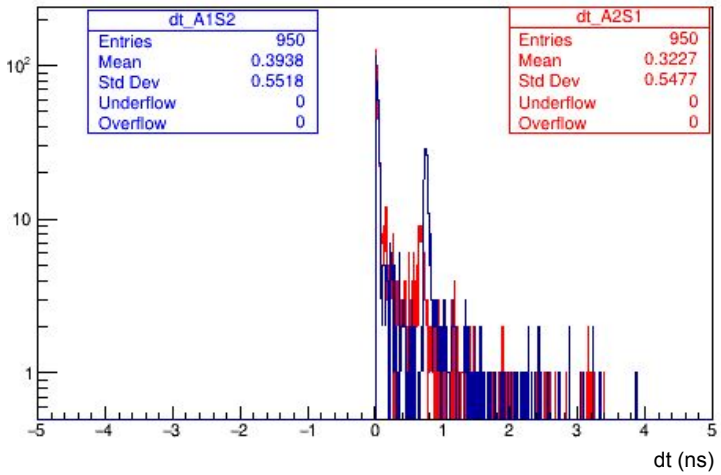
Time difference between actually not connected hits

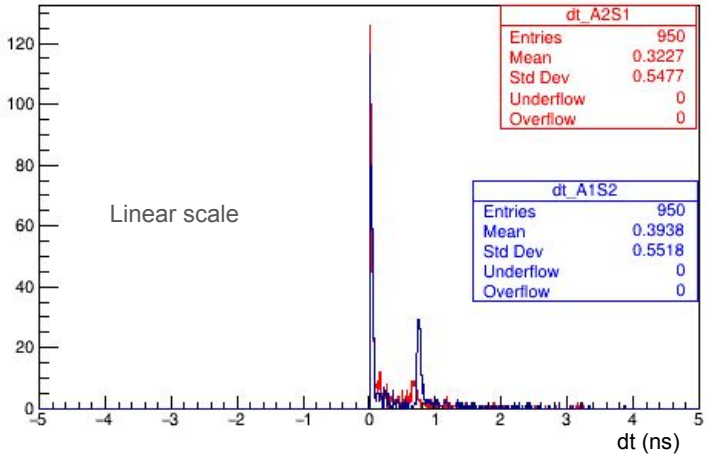




- There is some difference between, but I think they are expected.
- Some in A2S1, some events have dt (t_S1 - t_A2) -ve and +ve because it can happen that after scattered photon from A1 can make a hit before annihilation photon from second gun.
- Looking at the abosolute value of dt in the above case might be helpful.
- Don’t understand the reason for 2nd peak visible in the linear scale

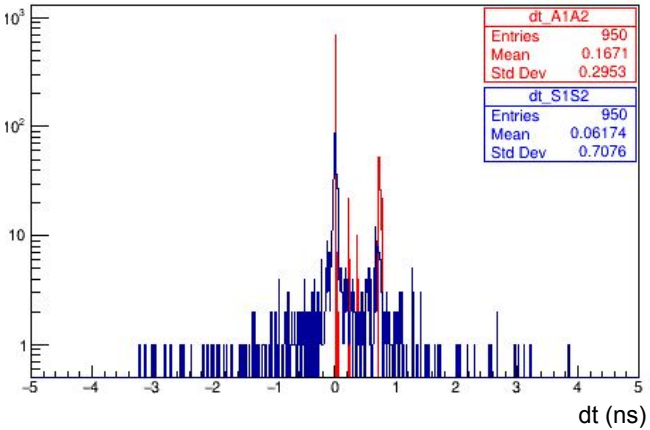
Looking at the absolute value of dt for the last slide

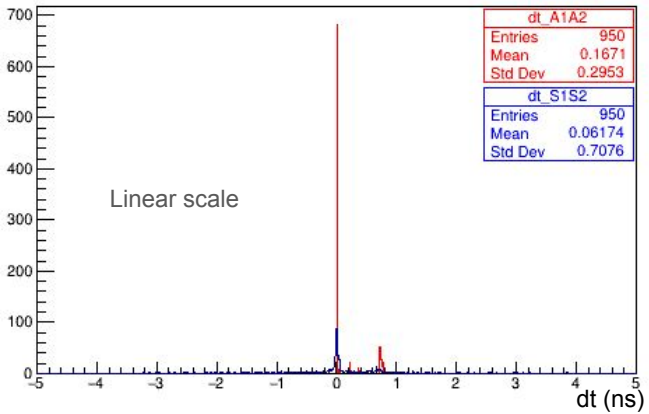




- Shape looks similar, but the height of the second peak is significantly different.

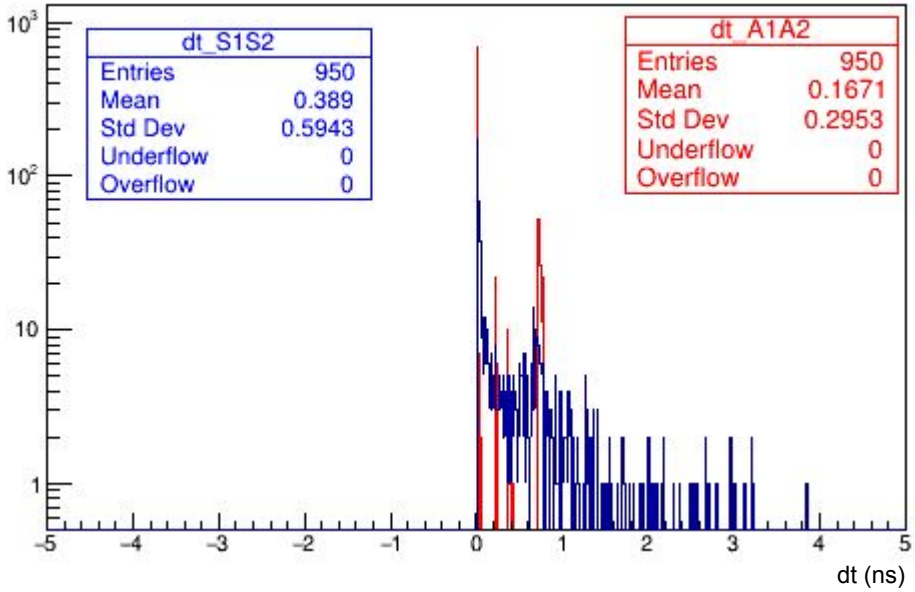
Comparing dt_A1A2 and S1S2





- dt is both negative and positive, because S1 can happen before or after S2. looking at the distribution of absolute value of dt might be helpful.
- But as compared to A1A2, S1S2 has less events at dt =0.
- Also an additional peak is visible in the linear scale. Which I am not sure why?

Plotting the absolute value of time in S1S2



The shape seems to be similar, when taken the absolute value of dt.